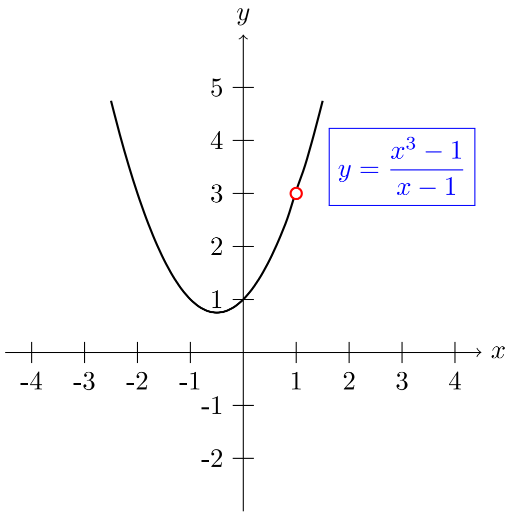
# Lecture Two

# 1 Limit of the Function

## 1.1 Introduction

Given the function $f(x) = \dfrac{x^3 - 1}{x - 1}$, clearly it is not defined at $x = 1$, and we want to study the behaviour of this function when $x$ approaches to 1.




The following table shows:

| $x$ | 0.75 | 0.9 | 0.99 | 0.999 |
|---|---|---|---|---|
| $f(x)$ | 2.313 | 2.710 | 2.970 | 2.997 |

**From the left**

| $x$ | 1.001 | 1.01 | 1.1 | 1.25 |
|---|---|---|---|---|
| $f(x)$ | 3.003 | 3.030 | 3.310 | 3.813 |

**From the right**

We can move arbitrarily close to 1 and as a result $f(x)$ moves arbitrarily close to 3, and here we can say the limit of the function $f(x) = \dfrac{x^3 - 1}{x - 1}$ is 3 when $x$ goes to 1. This limit is written as:

$$\lim_{x \to 1} \frac{x^3 - 1}{x - 1} = 3$$

**In general**: If $f(x)$ becomes arbitrarily close to a single number $L$ as $x$ approaches $c$ from either sides, the limit of $f(x)$ as $x$ approaches $c$ is $L$,and we can write it as:

$$\boxed{\lim_{x \to c} f(x) = L}$$

## 1.2 Definition of limit

Let $f$ be a function defined on an open interval containing $c$ (except possibly at $c$) and let $L$ be a real number. The statement $\lim_{x \to c} f(x) = L$ means that for each $\epsilon > 0$ there exist $\delta > 0$ such that if $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

**Example 1.2.1** *Prove that* $\lim_{x \to 3} (2x + 3) = 9$.

*Proof*

*We must show that for each* $\epsilon > 0$*, there exists a* $\delta > 0$ *such that* $|(2x+3)-9| < \epsilon$ *whenever* $0 < |x-3| < \delta$*, because our choice of* $\delta$ *depends on* $\epsilon$*. Now we need to establish a connection between the absolute value* $|(2x + 3) - 9|$ *and* $|x - 3|$.

$$|(2x + 3) - 9| = |2x - 6| = 2|x - 3|$$

*So for a given $\epsilon > 0$, we can choose $\delta = \frac{\epsilon}{2}$. This choice works because*

$$0 < |x-3| < \delta = \frac{\epsilon}{2}$$

*implies that*

$$|(2x+3)-9| = 2|x-3| < 2\left(\frac{\epsilon}{2}\right) = \epsilon.$$

## 1.3 One Sided Limit

1. **Left Hand Limit**: We mean by L-H-L, limit of the function when $x$ increasing to $c, (x \neq c)$, i.e. $(x < c)$ and $x \to c$ and we denote it by

$$\boxed{\lim_{x\to c^-} f(x)}$$

   To evaluate the left limit, we substitute $x = c - \epsilon$ on $f(x)$, then we take limit when $\epsilon \to 0$. $(\epsilon > 0)$

$$\lim_{x\to c^-} f(x) = \lim_{\epsilon\to 0} f(c-\epsilon)$$

2. **Right Hand Limit**: We mean by R-H-L, limit of the function when $x$ decreasing to $c, (x \neq c)$, i.e. $(x > c)$ and $x \to c$ and we denote it by

$$\boxed{\lim_{x\to c^+} f(x)}$$

   To evaluate the left limit, we substitute $x = c + \epsilon$ on $f(x)$, then we take limit when $\epsilon \to 0$. $(\epsilon > 0)$

$$\lim_{x\to c^+} f(x) = \lim_{\epsilon\to 0} f(c+\epsilon)$$

And the limit of the function **will be exists** if and only if:

$$\lim_{x\to c^+} f(x) = \lim_{x\to c^-} f(x)$$



**Example 1.3.2** *Find the limit $\lim_{x\to 0} \frac{|x|}{x}$ using L-H-L and R-H-L*

*Sol*

1. *L-H-L:*

$$\lim_{x\to 0^+} \frac{|x|}{x} = \lim_{\epsilon\to 0} \frac{|0-\epsilon|}{0-\epsilon} = \frac{|-\epsilon|}{-\epsilon} = \lim_{\epsilon\to 0} \frac{\epsilon}{-\epsilon}$$
$$= \lim_{\epsilon\to 0} -1 = -1.$$

2. *R-H-L:*

$$\lim_{x\to 0^+} \frac{|x|}{x} = \lim_{\epsilon\to 0} \frac{|0+\epsilon|}{0+\epsilon} = \frac{|\epsilon|}{\epsilon} = \lim_{\epsilon\to 0} \frac{\epsilon}{\epsilon}$$
$$= \lim_{\epsilon\to 0} 1 = 1.$$

*Since $\lim_{x\to 0^+} f(x) \neq \lim_{x\to 0^-} f(x)$, therefore the limit does not exist.*

**Example 1.3.3** *Find the limit $\lim_{x\to 1} \frac{x^2+x-2}{x-1}$ using L-H-L and R-H-L*

*Sol*

1. *L-H-L $\lim_{x\to 1^-} \frac{x^2+x-2}{x-1}$*

$$\lim_{x\to 1^-} \frac{x^2+x-2}{x-1} = \lim_{\epsilon\to 0} \frac{(1-\epsilon)^2+(1-\epsilon)-2}{(1-\epsilon)-1} = \lim_{\epsilon\to 0} \frac{1-2\epsilon+\epsilon^2+1-\epsilon-2}{-\epsilon} = \lim_{\epsilon\to 0} \frac{\epsilon(\epsilon-3)}{-\epsilon}$$
$$= \lim_{\epsilon\to 0}(3-\epsilon) = 3-0 = 3$$

2. *R-H-L:* $\lim_{x\to 1^+} \dfrac{x^2+x-2}{x-1}$

$$\lim_{x\to 1^+} \frac{x^2+x-2}{x-1} = \lim_{\epsilon\to 0} \frac{(1+\epsilon)^2+(1+\epsilon)-2}{(1+\epsilon)-1} = \lim_{\epsilon\to 0} \frac{1+2\epsilon+\epsilon^2+1+\epsilon-2}{\epsilon} = \lim_{\epsilon\to 0} \frac{\epsilon(\epsilon+3)}{\epsilon}$$
$$= \lim_{\epsilon\to 0}(\epsilon+3) = 0+3 = 3$$

*Since R-H-L = L-H-L, the limit exists and equal to 3.*

## 1.4 Properties of Limits

**Theorem 1.4.1** *Let $b$ and $c$ be real numbers and let $n$ be a positive integer.*

1. $$\lim_{x\to c} b = b$$
2. $$\lim_{x\to c} x = c$$
3. $$\lim_{x\to c} x^n = c^n$$

**Example 1.4.4** *Evaluate the limits*

1. $$\lim_{x\to 2} 3 = 3$$
2. $$\lim_{x\to -4} x = -4$$
3. $$\lim_{x\to 2} x^2 = (2)^2 = 4$$



**Theorem 1.4.2** *Let $b$ and $c$ be real numbers and let $n$ be a positive integer, and let $f$ and $g$ be functions with the following limits.*

$$\lim_{x\to c} f(x) = L \qquad , \qquad \lim_{x\to c} g(x) = K$$

*then*

1. *Scalar multiple*
$$\lim_{x\to c}[bf(x)] = b\lim_{x\to c} f(x) = bL$$
2. *Sum or Difference*
$$\lim_{x\to c}[f(x)\pm g(x)] = \lim_{x\to c} f(x) \pm \lim_{x\to c} g(x) = L\pm K$$
3. *Product*
$$\lim_{x\to c}[f(x).g(x)] = \lim_{x\to c} f(x). \lim_{x\to c} g(x) = L.K$$
4. *Quotient*
$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{\lim_{x\to c} f(x)}{\lim_{x\to c} g(x)} = \frac{L}{K} \qquad , K\neq 0$$
5. *Power*
$$\lim_{x\to c}[f(x)]^n = \left[\lim_{x\to c} f(x)\right]^n = L^n.$$

**Example 1.4.5** *By using theorems of limits evaluate the following limits.*

1. $\lim_{x\to 2}(4x^2+3)$

*Sol*

$$\begin{aligned}\lim_{x\to 2}(4x^2+3) &= \lim_{x\to 2}4x^2+\lim_{x\to 2}3 = 4\lim_{x\to 2}x^2+\lim_{x\to 2}3\\ &= 4\left(\lim_{x\to 2}x\right)^2+\lim_{x\to 2}3 = 4(2)^2+3 = 19.\end{aligned}$$

*2.* $\lim\limits_{x\to 3}\left[\dfrac{x-2}{x+2}\right]$

*Sol*

$$\lim_{x\to 3}\left[\frac{x-2}{x+2}\right] = \frac{\lim_{x\to 3}(x-2)}{\lim_{x\to 3}(x+2)} = \frac{\lim_{x\to 3}x-\lim_{x\to 3}2}{\lim_{x\to 3}x+\lim_{x\to 3}2} = \frac{3-2}{3+2} = \frac{1}{5}$$

*3.* $\lim\limits_{x\to 4}\sqrt{25-x^2}$

*Sol*

$$\lim_{x\to 4}(25-x^2)^{\frac{1}{2}} = \left(\lim_{x\to 4}25-\lim_{x\to 4}x^2\right)^{\frac{1}{2}} = \left(\lim_{x\to 4}25-\left(\lim_{x\to 4}x\right)^2\right)^{\frac{1}{2}} = \left(25-(4)^2\right)^{\frac{1}{2}} = 3$$

*4.* $\lim\limits_{x\to 1}(3x^2-1)(9x+2)$

**Example 1.4.6** *Evaluate the following limits:*

*1.* $\lim\limits_{x\to 1}(x+4-x^2)$

*Sol*

*2.* $\lim\limits_{x\to 3}\dfrac{3x+1}{x+2}$

*Sol*

*3.* $\lim\limits_{x\to 1}(x+4)(x-1)$

*Sol*

*4.* $\lim\limits_{x\to 2}\dfrac{x+2}{x-2}$

*Sol*

*5.* $\lim\limits_{x\to\infty} \dfrac{3+\dfrac{1}{x}}{2+x}$

*Sol*

*6.* $\lim\limits_{x\to\infty} \dfrac{4x+1}{x+9}$

*Sol*

*7.* $\lim\limits_{x\to-1} \dfrac{x^2-1}{x+1}$

*Sol*



*8.* $\lim\limits_{x\to-1} \dfrac{x^2+3x+2}{x+1}$

*Sol*

*9.* $\lim\limits_{x\to 2} \dfrac{x^5-32}{x^2-4}$

*Sol*

*10.* $\lim\limits_{x\to 0} \dfrac{x+\sin x}{\tan x + x}$

*Sol*

## 1.5 Strategies for Finding Limits

Previously, we studied several types of functions whose limits can be evaluated by **direct substitution**, this knowledge together with the following theorem can be used to develop a strategy for finding limits.

**Theorem 1.5.3** *Let $c$ be a real number and let $f(x) = g(x)$ for all $x \neq c$ in an open interval containing $c$. If the limit of $g(x)$ as $x$ approaches $c$ exists, then the limit of $f(x)$ also exists and:*

$$\lim_{x\to c} f(x) = \lim_{x\to c} g(x)$$

**Example 1.5.7** *Evaluate the following limits:*

1. $\displaystyle\lim_{x\to 2} \frac{x^2 - x - 2}{x^2 - 4}$

*Sol*

*By direct substitution, we get:*

$$\frac{(2)^2 - 2 - 2}{(2)^2 - 4} = \frac{4 - 2 - 2}{4 - 4} = \frac{0}{0} \qquad \textit{(undetermined value)}$$

*Now we try to find a function $g$ that agrees with $f$ for all $x$ other than $x = 2$. By factoring and dividing out like factors, we can rewrite the given function as:*

$$f(x) = \frac{x^2 - x - 2}{x^2 - 4} = \frac{\cancel{(x-2)}(x+1)}{\cancel{(x-2)}(x+2)} = \frac{x+1}{x+2} = g(x).$$

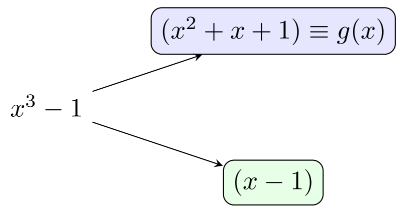
*So for all $x-$values other than $x = 2$ the functions $f$ and $g$ agree, since $\lim_{x\to 2} g(x)$ exists, we can apply theorem 1.5.3 to conclude that $f$ and $g$ have the same limit at $x = 2$.*

$$\lim_{x\to 2} \frac{x^2 - x - 2}{x^2 - 4} = \lim_{x\to 2} \frac{(x-2)(x+1)}{(x-2)(x+2)} = \lim_{x\to 2} \frac{x+1}{x+2} = \frac{2+1}{2+2} = \frac{3}{4}$$

2. $\displaystyle\lim_{x\to 1} \frac{x^3 - 1}{x - 1}$

*Sol*

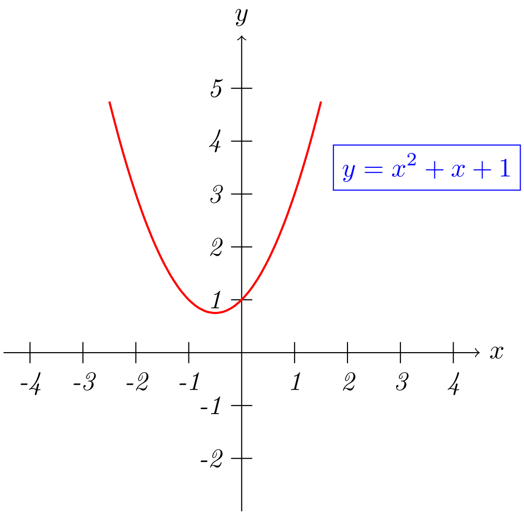
*By direct substitution we get $\dfrac{(1)^3 - 1}{1 - 1} = \dfrac{0}{0}$. Then we try to find a function $g$ that agrees with $f$ for all $x$ other than $x = 1$.*




$$f(x) = \frac{x^3 - 1}{x - 1} = \frac{(x-1)(x^2 + x + 1)}{(x-1)} = (x^2 + x + 1) = g(x)$$

*Thus $\lim_{x\to 1} f(x) = \lim_{x\to 1} g(x) = \lim_{x\to 1} (x^2 + x + 1) = 1^2 + 1 + 1 = 3.$*

***Note that:** the graph of the function $g(x) = x^2 + x + 1$ is same as $f(x) = \dfrac{x^3 - 1}{x - 1}$*




3. $\lim\limits_{x \to -1} \dfrac{x+1}{x^2+4x+3}$

*Sol*

$$\lim_{x \to -1} \frac{x+1}{x^2+4x+3} = \lim_{x \to -1} \frac{x+1}{(x+1)(x+3)} = \lim_{x \to -1} \frac{1}{(x+3)} = \frac{1}{-1+3} = \frac{1}{2}$$

4. $\lim\limits_{x \to 5} \dfrac{x-5}{\sqrt{x-3}-\sqrt{2}}$

*Sol*



$$\lim_{x \to 5} \frac{x-5}{\sqrt{x-3}-\sqrt{2}} . \frac{\sqrt{x-3}+\sqrt{2}}{\sqrt{x-3}+\sqrt{2}} = \lim_{x \to 5} \frac{(x-5)\left(\sqrt{x-3}+\sqrt{2}\right)}{(x-5)} = \lim_{x \to 5} \left(\sqrt{x-3}+\sqrt{2}\right) = \sqrt{2}+\sqrt{2} = 2\sqrt{2}.$$

5. $\lim\limits_{x \to 0} \dfrac{(1+x)^2-1}{x}$

*Sol*

6. $\lim\limits_{x \to 1} \dfrac{x^6-1}{x^2-1}$

*Sol*

7. $\lim\limits_{x \to 3} \dfrac{\frac{1}{x}-\frac{1}{3}}{x^2-9}$

*Sol*

8. $\lim\limits_{x\to 4} \dfrac{\sqrt{x}-2}{x-4}$

*Sol*

**Example 1.5.8** *Evaluate the following limits.*

1. $\lim\limits_{x\to\infty} \dfrac{9x^2}{2x^2+4}$

*Sol*

*By direct substitution, we get:*

$$\lim_{x\to\infty} \frac{9x^2}{2x^2+4} = \frac{9(\infty)^2}{2(\infty)^2+4} = \frac{\infty}{\infty} \qquad ,(\textit{undetermined value})$$

$$\lim_{x\to\infty} \underbrace{\frac{9x^2}{2x^2+4}}_{f(x)} = \lim_{x\to\infty} \left(\frac{\cancel{x^2}[9]}{\cancel{x^2}\left[2+\frac{4}{x^2}\right]}\right) = \lim_{x\to\infty} \underbrace{\left(\frac{9}{2+\frac{4}{x^2}}\right)}_{g(x)} = \frac{9}{2+\frac{4}{\infty^2}} = \frac{9}{2+0} = \frac{9}{2}$$

2. $\lim\limits_{x\to\infty} \dfrac{x^3-x^2+2}{2x^2+x+1}$

*Sol*

$$\lim_{x\to\infty} \underbrace{\frac{x^3-x^2+2}{2x^2+x+1}}_{f(x)} = \lim_{x\to\infty} \frac{\cancel{x^3}\left(1-\frac{1}{x}+\frac{2}{x^3}\right)}{\cancel{x^3}\left(\frac{2}{x}+\frac{1}{x^2}+\frac{1}{x^3}\right)} = \lim_{x\to\infty} \underbrace{\frac{\left(1-\frac{1}{x}+\frac{2}{x^3}\right)}{\left(\frac{2}{x}+\frac{1}{x^2}+\frac{1}{x^3}\right)}}_{g(x)} = \frac{\left(1-\frac{1}{\infty}+\frac{2}{\infty^3}\right)}{\left(\frac{2}{\infty}+\frac{1}{\infty^2}+\frac{1}{\infty^3}\right)} = \infty$$

3. $\lim\limits_{x\to\infty} \dfrac{x(x+1)}{x^3+1}$

*Sol*

*4.* $\lim\limits_{x\to\infty} \dfrac{6x+1}{\sqrt{4x^2+3}}$

*Sol*

*5.* $\lim\limits_{x\to\infty} \dfrac{(x-3)(2x+1)}{2x^2-x+1}$

*Sol*

*6.* $\lim\limits_{x\to\infty} \dfrac{(3x+4)(x-2)}{x(2x+1)(x+2)}$

*Sol*



*7.* $\lim\limits_{x\to\infty} \dfrac{\sqrt{9x^2+5}}{x+3}$

*Sol*

## 1.6 Cauchy's Theorem

**Theorem 1.6.4** ***Cauchy's Theorem:***

$$\boxed{\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}}$$

*proof*

*By using **long division**, we get:*

$$\frac{x^n - a^n}{x - a} = x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-1}$$

*Therefore:*

$$\begin{aligned}\lim_{x\to a}\frac{x^n - a^n}{x - a} &= \lim_{x\to a}\left(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-1}\right)\\ &= a^{n-1} + a\times a^{n-2} + a^2\times a^{n-3} + \cdots + a^{n-1}\\ &= \underbrace{a^{n-1} + a^{n-1} + a^{n-1} + \cdots + a^{n-1}}_{n\ times} = na^{n-1}\end{aligned}$$

**Lemma 1.6.1**

$$\boxed{\lim_{x\to a}\frac{x^n - a^n}{x^m - a^m} = \frac{n}{m}a^{n-m}}$$

*proof*

*We can write* $f(x) = \dfrac{x^n - a^n}{x^m - a^m} = \dfrac{x^n - a^n}{x - a} \div \dfrac{x^m - a^m}{x - a}$. *Then the limit will be:*

$$\lim_{x\to a}\frac{x^n - a^n}{x^m - a^m} = \lim_{x\to a}\frac{x^n - a^n}{x - a} \div \lim_{x\to a}\frac{x^m - a^m}{x - a} = \frac{na^{n-1}}{ma^{m-1}} = \frac{n}{m}a^{n-m}.$$

**Lemma 1.6.2**

$$\boxed{\lim_{x\to a}\frac{x - a}{x^m - a^m} = \frac{1}{m}a^{1-m}}$$

*Proof left for you.*

**Example 1.6.9** *Evaluate the following limits.*

1. $\displaystyle\lim_{x\to 2}\frac{x^5 - 32}{x^3 - 8}$

*Sol*

$$\lim_{x\to 2}\frac{x^5 - 32}{x^3 - 8} = \lim_{x\to 2}\frac{x^5 - (2)^5}{x^3 - (2)^3} = \frac{5}{3}(2)^{5-3} = \frac{5}{3}(2)^2 = \frac{20}{3}.$$

2. $\displaystyle\lim_{x\to 4}\frac{x^2 - 16}{x\sqrt{x} - 8}$

*Sol*

$$\lim_{x\to 4}\frac{x^2 - 16}{x\sqrt{x} - 8} = \lim_{x\to 4}\frac{x^2 - (4)^2}{x^{\frac{3}{2}} - (4)^{\frac{3}{2}}} = \frac{2}{\frac{3}{2}}(4)^{2-\frac{3}{2}} = \frac{4}{3}(2) = \frac{8}{3}$$

3. $\displaystyle\lim_{\Delta x\to 0}\frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x}$

*Sol*

$$\lim_{\Delta x\to 0}\frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x} = \lim_{x+\Delta x\to x}\frac{(x + \Delta x)^{\frac{1}{2}} - x^{\frac{1}{2}}}{(x + \Delta x) - x} = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

4. $\displaystyle\lim_{x\to 3}\frac{x - 3}{x^3 - 27}$

*Sol*

$$\lim_{x\to 3}\frac{x - 3}{x^3 - 27} = \lim_{x\to 3}\frac{x - 3}{x^3 - (3)^3} = \frac{1}{3}(3)^{1-3} = \frac{1}{3}.\frac{1}{9} = \frac{1}{27}.$$

5. $\displaystyle\lim_{x\to -1}\frac{x^3 + 1}{x + 1}$

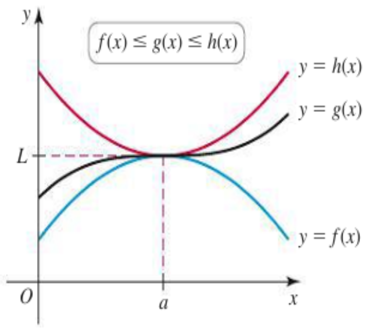
*Sol*

$$\lim_{x\to -1}\frac{x^3 + 1}{x + 1} = \lim_{x\to -1}\frac{x^3 - (-1)^3}{x - (-1)} = 3(-1)^{3-1} = 3(-1)^2 = 3.$$

## 1.7 The squeeze Theorem

**Theorem 1.7.5** *Assume the functions $f, g$ and $h$ satisfy $f(x) \leq g(x) \leq h(x)$ for all values of $x$ near $a$, except possibly at $a$.*

$$\boxed{If \ \lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L, \ then \ \lim_{x\to a} g(x) = L}$$




## 1.8 Limit of Trigonometric Functions

We have some important limits:

1. $\boxed{\lim_{x\to 0} \sin x = 0}$
2. $\boxed{\lim_{x\to 0} \cos x = 1}$
3. $\boxed{\lim_{x\to 0} \frac{\sin x}{x} = 1}$
4. $\boxed{\lim_{ax\to 0} \frac{\sin ax}{ax} = 1}$
5. $\boxed{\lim_{x\to 0} \frac{\tan x}{x} = 1}$
6. $\boxed{\lim_{ax\to 0} \frac{\tan ax}{ax} = 1}$

**Example 1.8.10** *1. Prove that* $\boxed{\lim_{x\to 0} \frac{\tan x}{x} = 1}$

*Sol*

$$\lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \frac{\sin x}{x\cos x} = \lim_{x\to 0} \frac{\sin x}{x}. \lim_{x\to 0} \frac{1}{\cos x} = 1 \times \frac{1}{\cos 0} = 1 \times 1 = 1$$

*2.* $\lim_{x\to 0} \frac{\sin 2x}{x}$

*Sol*

$$\lim_{x\to 0} \frac{\sin 2x}{x} = \lim_{2x\to 0} \frac{2\sin 2x}{2x} = 2 \lim_{2x\to 0} \frac{\sin 2x}{2x} = 2 \times 1 = 2$$

*3.* $\lim_{x\to 0} \frac{\tan 3x}{2x}$

*Sol*

$$\lim_{x\to 0} \frac{\tan 3x}{2x} = \frac{1}{2} \lim_{x\to 0} \frac{\tan 3x}{x} = \frac{1}{2} \lim_{3x\to 0} \frac{3\tan 3x}{3x} = \frac{3}{2} \lim_{3x\to 0} \frac{\tan 3x}{3x} = \frac{3}{2} \times 1 = \frac{3}{2}.$$

4. $\lim\limits_{x\to 0} \dfrac{\tan x + \sin 3x}{x}$

*Sol*

$$\lim_{x\to 0} \frac{\tan x + \sin 3x}{x} = \lim_{x\to 0} \frac{\tan x}{x} + \lim_{x\to 0} \frac{\sin 3x}{x} = \lim_{x\to 0} \frac{\tan x}{x} + 3 \lim_{3x\to 0} \frac{\sin 3x}{3x} = 1 + 3 \times 1 = 4.$$

5. $\lim\limits_{x\to 0} \dfrac{2x + \sin x}{4x + \tan 5x}$

*Sol*

$$\begin{aligned}
\lim_{x\to 0} \frac{2x + \sin x}{4x + \tan 5x} &= \lim_{x\to 0} \left( \frac{\dfrac{2x + \sin x}{x}}{\dfrac{4x + \tan 5x}{x}} \right) = \lim_{x\to 0} \frac{2x + \sin x}{x} \div \lim_{x\to 0} \frac{4x + \tan 5x}{x} \\
&= \left( \lim_{x\to 0} 2 + \lim_{x\to 0} \frac{\sin x}{x} \right) \div \left( \lim_{x\to 0} 4 + \lim_{x\to 0} \frac{\tan 5x}{x} \right) \\
&= \left( \lim_{x\to 0} 2 + \lim_{x\to 0} \frac{\sin x}{x} \right) \div \left( \lim_{x\to 0} 4 + 5 \lim_{5x\to 0} \frac{\tan 5x}{5x} \right) \\
&= \frac{2+1}{4+5} = \frac{3}{9} = \frac{1}{3}.
\end{aligned}$$

6. $\lim\limits_{x\to \infty} x \sin\left(\dfrac{3}{x}\right)$

*Sol*

*Let* $u = \dfrac{3}{x} \Rightarrow x = \dfrac{3}{u}$ *and* $u \to 0$ *as* $x \to \infty$. *Thus*

$$\lim_{x\to \infty} x \sin\left(\frac{3}{x}\right) = \lim_{u\to 0} \frac{3}{u} \sin u = 3 \lim_{u\to 0} \frac{\sin u}{u} = 3 \times 1 = 3.$$



7. $\lim\limits_{x\to 0} \dfrac{1 - \cos x}{x^2}$

*Sol*

$$\begin{aligned}
\lim_{x\to 0} \frac{1 - \cos x}{x^2} \times \frac{1 + \cos x}{1 + \cos x} &= \lim_{x\to 0} \frac{1 - \cos^2 x}{x^2(1 + \cos x)} = \lim_{x\to 0} \frac{\sin^2 x}{x^2(1 + \cos x)} = \lim_{x\to 0} \frac{\sin^2 x}{x^2} \times \lim_{x\to 0} \frac{1}{(1 + \cos x)} \\
&= \left( \lim_{x\to 0} \frac{\sin x}{x} \right)^2 \times \lim_{x\to 0} \frac{1}{(1 + \cos x)} = 1 \times \frac{1}{(1 + \cos 0)} = \frac{1}{2}
\end{aligned}$$

## 1.9 The Number e

We define the number $e$ by the limit

$$e = \lim_{x\to \infty} \left(1 + \frac{1}{x}\right)^x$$

Proof

$$\begin{aligned}
\left(1 + \frac{1}{x}\right)^x &= 1 + \binom{x}{1} . \frac{1}{x} + \binom{x}{2} . \frac{1}{x^2} + \binom{x}{3} . \frac{1}{x^3} + \cdots + \binom{x}{n} . \frac{1}{x^n} \\
&= \sum_{n=0}^{x} \frac{1}{n!} \left(1 - \frac{1}{x}\right) . \left(1 - \frac{2}{x}\right) . \left(1 - \frac{3}{x}\right) \ldots \left(1 - \frac{n-1}{x}\right)
\end{aligned}$$

Therefore the limit will be:

$$\begin{aligned}
\lim_{x\to \infty} \left(1 + \frac{1}{x}\right)^x &= \sum_{n=0}^{x} \left( \lim_{x\to \infty} \frac{1}{n!} \left(1 - \frac{1}{x}\right) . \left(1 - \frac{2}{x}\right) . \left(1 - \frac{3}{x}\right) \ldots \left(1 - \frac{n-1}{x}\right) \right) \\
&= = \sum_{n=0}^{x} \frac{1}{n!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \ldots
\end{aligned}$$

The number $e$ satisfied the inequality $2 < e < 3$, and its value $\approxeq 2.71828$, and we can write the limit as:

$$\boxed{e = \lim_{n\to \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n\to 0} (1 + n)^{\frac{1}{n}}}$$

## 1.10 The Exponential Theorem

**Theorem 1.10.6**

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

*Proof*

*We know that* $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$.

$$\Rightarrow e^x = \left[\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n\right]^x = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{nx}$$

*put* $m = nx \Rightarrow e^x = \lim_{m\to\infty} \left(1 + \frac{x}{m}\right)^m$, *therefore the limit will be:*

$$\begin{aligned}
&\Rightarrow \lim_{m\to\infty} \left[1 + \frac{mx}{m} + \frac{m(m-1)}{2!}\frac{x^2}{m^2} + \frac{m(m-1)(m-2)}{3!}\frac{x^3}{m^3} + \dots\right] \\
&= \lim_{m\to\infty} \left[1 + x + \frac{\left(1 - \frac{1}{m}\right)}{2!} + \frac{\left(1 - \frac{1}{m}\right)\left(1 - \frac{2}{m}\right)}{3!} + \dots\right] \\
&= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots
\end{aligned}$$

**Example 1.10.11** *1.* $\lim_{x\to 0} (1+x)^{\frac{1}{x}}$

*Sol*

*put* $u = \frac{1}{x} \Rightarrow x = \frac{1}{u}$ *and* $u \to \infty$ *as* $x \to 0$

$$\lim_{x\to 0} (1+x)^{\frac{1}{x}} = \lim_{u\to\infty} \left(1 + \frac{1}{u}\right)^u = e.$$

*2.* $\lim_{x\to\infty} \left(1 + \frac{\alpha}{x}\right)^x \qquad \alpha \neq 0$

*Sol*

*(i) If* $\alpha > 0$, *let* $x = \alpha u$

$$\lim_{x\to\infty} \left(1 + \frac{\alpha}{x}\right)^x = \lim_{u\to\infty} \left(1 + \frac{\alpha}{\alpha u}\right)^{\alpha u} = \left(\lim_{u\to\infty} \left(1 + \frac{1}{u}\right)^u\right)^\alpha = e^\alpha.$$

*(ii) If* $\alpha < 0 \Rightarrow$ *let* $x = -\alpha u$

$$\begin{aligned}
\lim_{x\to\infty} \left(1 + \frac{\alpha}{x}\right)^x &= \lim_{u\to\infty} \left(1 - \frac{\alpha}{\alpha u}\right)^{-\alpha u} = \left(\lim_{u\to\infty} \left(1 - \frac{1}{u}\right)^{-u}\right)^\alpha = \left(\lim_{u\to\infty} \left(\frac{u}{u-1}\right)^u\right)^\alpha \\
&= \left[\lim_{u\to\infty} \left(1 + \frac{1}{u-1}\right)^u\right]^\alpha
\end{aligned}$$

*Now let* $u - 1 = y \Rightarrow u = 1 + y$ $(y \to \infty$ *as* $u \to \infty$.

$$\begin{aligned}
\lim_{x\to\infty} \left(1 + \frac{\alpha}{x}\right)^x &= \left[\lim_{u\to\infty} \left(1 + \frac{1}{u-1}\right)^u\right]^\alpha = \left[\lim_{y\to\infty} \left(1 + \frac{1}{y}\right)^{y+1}\right]^\alpha \\
&= \left[\lim_{y\to\infty} \left(1 + \frac{1}{y}\right)^y \times \lim_{y\to\infty} \left(1 + \frac{1}{y}\right)\right]^\alpha = (e \times 1)^\alpha = e^\alpha.
\end{aligned}$$

*3.* $\lim_{x\to\infty}\left(\frac{x+3}{x-2}\right)^x$

*Sol*

*Let* $u = x-2 \Rightarrow x = u+2 \Rightarrow (u\to\infty)$ *as* $(x\to\infty)$

$$\Rightarrow \lim_{x\to\infty}\left(\frac{x+3}{x-2}\right)^x = \lim_{u\to\infty}\left(\frac{u+2+3}{u}\right)^{u+2} = \lim_{u\to\infty}\left(1+\frac{5}{u}\right)^{u+2} = \lim_{u\to\infty}\left(1+\frac{5}{u}\right)^{u}.\lim_{u\to\infty}\left(1+\frac{5}{u}\right)^{2} = e^5$$

**Exercise 1.10.1** *Evaluate the following limits.*

*1.* $\lim_{x\to3}\frac{x^2-2x-3}{x-3}$

*Sol*

*2.* $\lim_{x\to4}\frac{x^2-16}{4-x}$

*Sol*



*3.* $\lim_{x\to b}\frac{(x-b)^{50}-x+b}{x-b}$

*Sol*

*4.* $\lim_{h\to0}\frac{\frac{1}{5+h}-\frac{1}{5}}{h}$

*Sol*

*5.* $\lim_{w\to1}\left(\frac{1}{w^2-w}-\frac{1}{w-1}\right)$

*Sol*

6. $\lim_{x\to\infty} \dfrac{x^2+2x+9}{3x^3+x+1}$

*Sol*

7. $\lim_{x\to 1} \dfrac{\sqrt{9x^2+7}}{3x+1}$

*Sol*

8. $\lim_{n\to\infty} \left(\dfrac{1+2+3+\cdots+n}{n^2}\right)$

*Sol*

9. $\lim_{x\to 1} \dfrac{x^3+x^2+5x-5}{x-1}$



*Sol*

10. $\lim_{x\to 0} \left(\dfrac{\dfrac{1}{x-1}+\dfrac{1}{x+1}}{x}\right)$

*Sol*

11. $\lim_{x\to 2} \dfrac{x^3(x-1)^6-8}{x-2}$

*Sol*

12. $\lim_{x\to 2} \dfrac{\sqrt{x^2+12}-4}{x-2}$

*Sol*

13. $\lim\limits_{x \to 1} \dfrac{x^{-1} - 1}{x - 1}$

*Sol*

14. $\lim\limits_{x \to 4} \dfrac{4x^2 - x^2}{2 - \sqrt{x}}$

*Sol*

15. $\lim\limits_{x \to 4} \dfrac{4 - x}{\sqrt{x^2 + 9} - 5}$

*Sol*

16. $\lim\limits_{y \to 8} \dfrac{\sqrt[3]{y} - 2}{y\sqrt[3]{y} - 16}$

*Sol*

17. $\lim\limits_{y \to 3} \dfrac{y^4 - 81}{y^2 - 9}$

*Sol*

18. $\lim\limits_{\Delta x \to 0} \dfrac{(x + \Delta x)^3 - x^3}{\Delta x}$

*Sol*

19. $\lim\limits_{x \to -2} \dfrac{x^5 + 32}{x^3 + 8}$

*Sol*

20. $\lim\limits_{x \to 0} \dfrac{4x + \sin 3x}{x}$

*Sol*

21. $\lim\limits_{x \to 0} \dfrac{\sin 3x}{\sin 9x}$

*Sol*

22. $\lim\limits_{x \to 0} \dfrac{\sin 2x}{x^2 + 3x}$



*Sol*

23. $\lim\limits_{x \to 2} \dfrac{\sin(x - 2)}{x^2 - 4}$

*Sol*

24. $\lim\limits_{x \to 0} \dfrac{1 - \cos 4x}{x^2}$

*Sol*

25. $\lim\limits_{x \to 0} \dfrac{\tan 2x + 4x}{x + \sin x + \tan x}$

*Sol*

*26.* $\lim_{x \to 0} \frac{\cos 4x - \cos 2x}{x^2}$

*Sol*

*27.* $\lim_{x \to \infty} x \sin \frac{6}{x}$

*Sol*

# Lecture Three

## 1 The Continuity

### 1.1 Continuity at a point on an open interval

**Informally**, we might say that a function is continuous on an open interval if its graph can be drawn with a pencil without lifting the pencil from the paper. i.e. A function $f$ is continuous at $x = c$ means there is no interruption in the graph of $f$ at $c$. That is, its graph is unbroken at $c$ and there are no holes, jumps, or gaps. The following figure identifies three values of $x$ at which the graph of $f$ is not continuous. At all other points in the interval $(a, b)$ the graph of $f$ is uninterrupted and **continuous**.

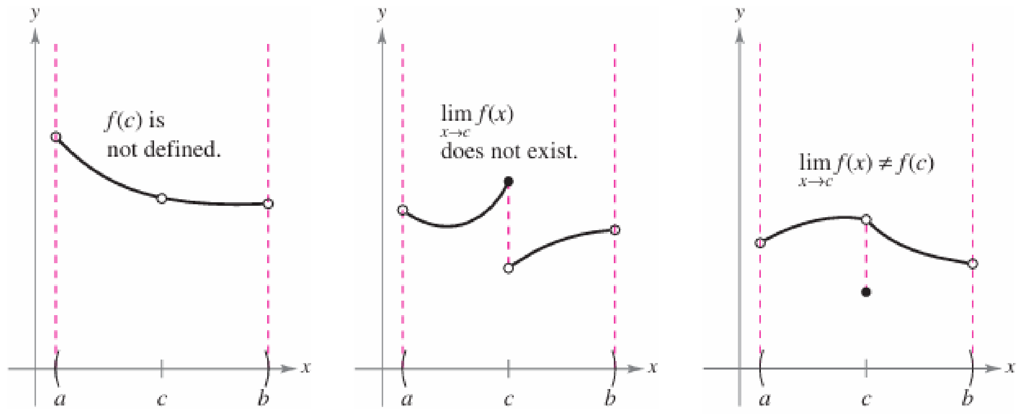



Figure 1: Three cases for the discontinuity



### 1.2 Concept of Continuity

**Definition 1.2.1** ***Continuity at a point:***
*A function $f$ is continuous at $c$ if the following three conditions are met.*

1. *$f(c)$ is defined*
2. *$\lim_{x\to c} f(x)$ exists.*
3. *$f(c) = \lim_{x\to c} f(x)$*

***Continuity on an open interval:***
*A function is **continuous on an open interval** $(a, b)$ if it is continuous at each point in the interval.*

**Definition 1.2.2** *Let $f(x)$ be a **defined function** at $x = c$, we say that $f(x)$ is continuous at $x = c$ if the following condition satisfied.*

$$\boxed{\lim_{h\to 0}\,[f(c+h) - f(c)] = 0}$$

**Example 1.2.1** *Given $f(x) = x^2$, prove that it is continuous on $\mathbb{R}$.*

*Proof*

*To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c \in \mathbb{R}$.*

$$\begin{aligned}\lim_{h\to 0}\,[f(c+h) - f(c)] &= \lim_{h\to 0}\left[f(c+h)^2 - c^2\right] = \lim_{h\to 0}\left[c^2 + 2ch + h^2 - c^2\right] \\ &= \lim_{h\to 0}\,[h(2c+h)] = 0 \times (2c+0) = 0.\end{aligned}$$

*$\Rightarrow f(x) = x^2$ is continuous at $c \Rightarrow$ continuous at $\mathbb{R}$.*

**Example 1.2.2** *Prove that the function $f(x) = \cos ax$, continuous on $\mathbb{R}$.*

*Proof*

*To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c \in \mathbb{R}$.*

$$\begin{aligned}\lim_{h\to 0}[f(c+h)-f(c)] =& \lim_{h\to 0}[\cos(ac+ah)-\cos(ac)] = -2\lim_{h\to 0}\left[\sin\left(\frac{ac+ah+ac}{2}\right)\sin\left(\frac{ac+ah-ac}{2}\right)\right]\\ =& -2\lim_{h\to 0}\left[\sin\left(ac+\frac{ah}{2}\right)\sin\left(\frac{ah}{2}\right)\right] = -2\times\sin ac\times 0 = 0.\end{aligned}$$

$\Rightarrow f(x) = \cos ax$ *is continuous at* $c \Rightarrow$ *continuous at* $\mathbb{R}$.

**Example 1.2.3** *Prove that the function $f(x) = a^x$ , $a > 0$ is continuous on $\mathbb{R}$.*

*Proof*

*To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c \in \mathbb{R}$.*

$$\begin{aligned}\lim_{h\to 0}[f(c+h)-f(c)] =& \lim_{h\to 0}\left[a^{c+h}-a^c\right] = \lim_{h\to 0}\left[a^c\left(a^h-1\right)\right]\\ =& a^c(1-1) = a^c\times 0 = 0\end{aligned}$$

$\Rightarrow f(x) = a^x$ *is continuous on* $\mathbb{R}$

**Example 1.2.4** *Prove that the function $f(x) = \log_a x$ , $a > 0$ is continuous on $\mathbb{R}^+$.*

*Proof*

*To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c > 0, c \in \mathbb{R}$.*

$$\begin{aligned}\lim_{h\to 0}[f(c+h)-f(c)] =& \lim_{h\to 0}[\log_a(c+h)-\log_a c] = \lim_{h\to 0}\left[\log_a\left(\frac{c+h}{c}\right)\right]\\ =& \lim_{h\to 0}\left[\log_a\left(1+\frac{h}{c}\right)\right] = \log_a\left(\lim_{h\to 0}\left[1+\frac{h}{c}\right]\right) = \log_a(1) = 0\end{aligned}$$

**Example 1.2.5** *Discuss the continuity of the function*

$$f(x) = \begin{cases} 3x-1 & ,x<1\\ 2x^2 & ,x\ge 1\end{cases}$$

*at $x = 1$.*

*Sol*

*Here we will study the three conditions on definition 1.2.1.*

1. $f(1) = 2(1)^2 = 2$
2. *For finding the limit:*

$$\begin{aligned}\lim_{x\to 1^+} f(x) =& \lim_{\epsilon\to 0} f(a+\epsilon) = \lim_{\epsilon\to 0} 2(1+\epsilon)^2 = 2(1+0)^2 = 2(1)^2 = 2.\\ \lim_{x\to 1^-} f(x) =& \lim_{\epsilon\to 0} f(a-\epsilon) = \lim_{\epsilon\to 0} 3(1-\epsilon)-1 = 3(1)-1 = 3-1 = 2.\end{aligned}$$

   *Since* $\lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x) \Rightarrow \lim_{x\to 1} f(x)$ *exists and* $= 2$

3. *Clearly $f(1) = \lim_{x\to 1} f(x)$, therefore the given function is continuous at $x = 1$.*

**Example 1.2.6** *Determine whether the following function is continuous at $x = 0$.*

$$f(x) = \begin{cases} \dfrac{\sin \pi x}{x} & ,x<0\\ \dfrac{\sin \pi x}{x+1} & ,x\ge 0\end{cases}$$

*Sol*

1. $f(1) = \dfrac{\sin(0\pi)}{0+1} = \dfrac{0}{1} = 0$

2.

$$\lim_{x\to 0^+} f(x) = \lim_{\epsilon\to 0} f(a+\epsilon) = \lim_{\epsilon\to 0} \frac{\sin(\pi(0+\epsilon))}{0+\epsilon+1} = \lim_{\epsilon\to 0} \frac{\sin \pi\epsilon}{\epsilon+1} = \frac{\sin 0}{0+1} = \frac{0}{1} = 0$$

$$\lim_{x\to 0^-} f(x) = \lim_{\epsilon\to 0} f(a-\epsilon) = \lim_{\epsilon\to 0} \frac{\sin(\pi(0-\epsilon))}{0-\epsilon} = \lim_{\epsilon\to 0} \frac{\sin \pi\epsilon}{\epsilon} = \pi$$

$\Rightarrow \lim_{x\to 0} f(x)$ *not exist because* $\lim_{x\to 0^+} f(x) \neq \lim_{x\to 0^-} f(x)$

**Example 1.2.7** *Find the constant such that the function is continuous on the entire real line.*

1.

$$f(x) = \begin{cases} 3x^2 & , x \geq 1 \\ ax-4 & , x < 1 \end{cases}$$

2.

$$f(x) = \begin{cases} \dfrac{4\sin x}{x} & , x < 0 \\ a-2x & , x \geq 0 \end{cases}$$



3.

$$f(x) = \begin{cases} 1-4x & , x < 2 \\ ax^2-3x+2 & , x \geq 2 \end{cases}$$

## 1.3 Properties of Continuity

**Theorem 1.3.1** *If $b$ is a real number and $f$ and $g$ are continuous at $x = c$, then the following functions are also continuous at $c$*

1. *Scalar multiple:* $bf$
2. *Sum or Difference:* $f \pm g$
3. *Product:* $fg$
4. *Quotient:* $\dfrac{f}{g}$ *, if* $g(c) \neq 0$.

**Note:** The following types of functions are continuous at every point in their domains.

(i) Polynomials

(ii) Rational $r(x) = \dfrac{p(x)}{q(x)} \qquad , q(x) \neq 0$

(iii) Radical: $f(x) = \sqrt[n]{x}$

(iv) Trigonometric: $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$

**Exercise 1.3.1** *Find the following:*

1. *Prove that* $f(x) = \sin(ax) \quad , (a \neq 0)$ *is continuous on* $\mathbb{R}$.

2. *Prove that* $f(x) = x^3 - x$ *is continuous on* $\mathbb{R}$.

3. *Discuss the continuity of the function* $f(x) = \ln(x^2 + 1)$

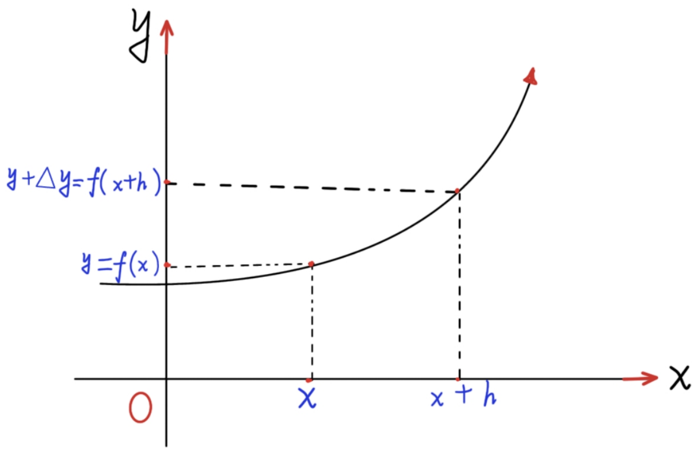
4. *Determine whether the following functions are continuous at the given points.*

   (a) $f(x) = \begin{cases} x^x - 1 & , x \neq 0 \\ 1 & , x = 0 \end{cases}$ *at* $x = 0$

   (b) $f(x) = \begin{cases} \dfrac{sin(\pi x)}{1 - x} & , x < 1 \\ \pi x^2 & , x \geq 1 \end{cases}$ *at* $x = 1$

# 2 The Differentiation

## 2.1 The Change and Rate of Change




Let $y = f(x)$ is a continuous function on an open interval $I$, $(x \in I)$.

1. The change on $x \equiv \Delta x = x_2 - x_1 = h$
2. The change on $y \equiv \Delta y = f(x+h) - f(x)$
3. The rate of change is $\dfrac{\Delta y}{\Delta x}$

$$\boxed{\frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h}}$$



**Example 2.1.8** *If $y = x^3$, find $\dfrac{\Delta y}{h}$*

*Sol*

$$\frac{\Delta y}{h} = \frac{(x+h)^3 - x^3}{h} = \frac{h\left[(x+h)^2 + x(x+h) + x^2\right]}{h}$$
$$= (x+h)^2 + x(x+h) + x^2 = 3x^2 + 3hx + h^2$$

*Therefore* $\dfrac{\Delta y}{h} = \boxed{3x^2 + 3hx + h^2}$

## 2.2 Rate of Changes or Derivatives

**Definition 2.2.3** *The derivative of $f$ is the function:*

$$f^{'}(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{\Delta y}{h} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{2.2.1}$$

*provided the limit exists and $x$ is in the domain of $f$. If $f^{'}(x)$ exists, we say that $f$ is **differentiable** at $x$.*

**Note:** If $f$ is differentiable at every point of an open interval $I$, we say that $f$ is differentiable on $I$.

**Example 2.2.9** *Given $y = x^3$, evaluate the rate of change $\dfrac{dy}{dx}$.*

*Sol*

*Since $y = x^3$ is a polynomial $\Rightarrow$ continuous on $\mathbb{R}$. By using 2.2.1*

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

*from example 2.1.8, we saw that* $\dfrac{f(x+h)-f(x)}{h} = 3x^2 + 3hx + h^2$

$$L.S.L = \lim_{h\to 0^+}(3x^2+3hx+h^2) = \lim_{h\to 0}(3x^2+3hx+h^2) = 3x^2.$$
$$R.S.L = \lim_{h\to 0^-}(3x^2+3hx+h^2) = \lim_{h\to 0}(3x^2+3hx+h^2) = 3x^2$$

*Therefore* $\boxed{\dfrac{dy}{dx} = 3x^2}$

**Example 2.2.10** *For the function $y = |x|$, find $\dfrac{dy}{dx}$ at $x = 0$.*

*Sol*

*We know that, the given function is continuous on $\mathbb{R}$.*

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{f(h)-0}{h} = \lim_{h\to 0}\frac{f(h)}{h}$$

*Now, we will study the existence of the limit by calculating the left and right limits.*

(i) $L.S.L = \lim\limits_{h\to 0^-}\dfrac{|h|}{h} = \lim\limits_{h\to 0}\dfrac{h}{-h} = -1$

(ii) $L.S.L = \lim\limits_{h\to 0^+}\dfrac{|h|}{h} = \lim\limits_{h\to 0}\dfrac{h}{h} = 1$

*Thus the limit does not exist at $x = 0$ and here we say the function is not differentiable at $x = 0$.*

**Theorem 2.2.2** ***Differentiable implies continuous***

*Proof*

*Because $f$ is differentiable at a point $c$, we know that:*

$$f^{'}(c) = \lim_{x\to c}\frac{f(x)-f(c)}{x-c}$$

*exists. To show that $f$ is continuous at $c$, we mush show that $\lim\limits_{x\to c} f(x) = f(c)$.*
*We can write $f(x)$ as:*

$$f(x) = \frac{f(x)-f(c)}{x-c}(x-c) + f(c) \qquad ,x \neq c \tag{2.2.2}$$

*By taking the limit as $x$ approaches $c$ on both sides of equation 2.2.2 and simplifying, we get:*

$$\begin{aligned}\lim_{x\to c} f(x) &= \lim_{x\to c}\left[\frac{f(x)-f(c)}{x-c}(x-c)+f(c)\right]\\ &= \underbrace{\lim_{x\to c}\frac{f(x)-f(c)}{x-c}}_{f^{'}(c)} \times \underbrace{\lim_{x\to c}(x-c)}_{0} + \underbrace{\lim_{x\to c} f(c)}_{f(c)} = f^{'}(c)\times 0 + f(c) = f(c)\end{aligned}$$

*Therefore $\lim\limits_{x\to c} f(x) = f(c)$ which means that $f$ is continuous at $x$.*

**Theorem 2.2.3** ***Not continuous implies not differentiable.***
*If $f$ is not continuous at $x = c$, then $f$ is not differentiable at $x = c$*

## 2.3 Rules of Differentiation

1. **The Constant function** $y = c$.

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{c-c}{h} = \lim_{h\to 0}\frac{0}{h} = 0$$

$$\boxed{\therefore \frac{d}{dx}[c] = 0}$$

2. **Power Rule**

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

Proof

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h\to 0}\frac{(x+h)^n - x^n}{h} = \lim_{h\to 0}\frac{\left[x^n + nx^{n-1}h + \binom{n}{2}n^{n-2}h^2 + \cdots + h^n\right] - x^n}{h} \\
&= \lim_{h\to 0}\frac{h\left[nx^{n-1} + \binom{n}{2}n^{n-1}h + \cdots + h^{n-1}\right]}{h} \\
&= \lim_{h\to 0}\left[nx^{n-1} + \binom{n}{2}n^{n-1}h + \cdots + h^{n-1}\right] = nx^{n-1}.
\end{aligned}$$

3. **The Sine function** $y = \sin x$

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h\to 0}\frac{\sin(x+h) - \sin x}{h} = \lim_{h\to 0}\frac{2\cos\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h} \\
&= \left[\lim_{h\to 0}\cos\left(x+\frac{h}{2}\right)\right]\left[\lim_{\frac{h}{2}\to 0}\frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)}\right] = \cos x \times 1 = \cos x.
\end{aligned}$$

$$\boxed{\therefore \frac{d}{dx}[\sin x] = \cos x}$$

4. **The Cosine function** $y = \cos x$

$$\frac{d}{dx}[\cos x] = -\sin x$$

Proof, left for you.

5. **The Tangent Function** $y = \tan x$

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h\to 0}\frac{\tan(x+h) - \tan x}{h} = \lim_{h\to 0}\frac{\frac{\tan x + \tan h}{1 - \tan x\tan h} - \tan x}{h} \\
&= \lim_{h\to 0}\frac{\tan x + \tan h - \tan x(1 - \tan x\tan h)}{h(1 - \tan x\tan h)} \\
&= \lim_{h\to 0}\frac{\cancel{\tan x} + \tan h - \cancel{\tan x} + \tan^2 x\tan h}{h(1 - \tan x\tan h)} \\
&= \underbrace{\lim_{h\to 0}\left(\frac{\tan h}{h}\right)}_{(1)} \times \underbrace{\lim_{h\to 0}\left(\frac{1}{1 - \tan x\tan h}\right)}_{(1)} \times \underbrace{\lim_{h\to 0}(1 + \tan^2 x)}_{\sec^2 x} = 1 \times 1 \times \sec^2 x = \sec^2 x
\end{aligned}$$

6. **The Exponential function** $y = a^x \quad , a > 0\ (a \neq 1)$.

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{a^{x+h} - a^x}{h} = a^x \lim_{h\to 0}\frac{a^h - 1}{h}$$

Now, let $\frac{1}{u} = a^h - 1 \Rightarrow u = \frac{1}{a^h - 1}$, thus $u \to \infty$ when $h \to 0$ and $h = \log_a\left(1 + \frac{1}{u}\right)$

$$\begin{aligned}
\Rightarrow \frac{dy}{dx} &= a^x \times \lim_{u\to\infty}\frac{1}{u\left(\log_a\left(1+\frac{1}{u}\right)\right)} = a^x \times \lim_{u\to\infty}\frac{1}{\log_a\left(1+\frac{1}{u}\right)^u} \\
&= a^x \times \frac{1}{\log_a\left(\lim_{u\to\infty}\left(1+\frac{1}{u}\right)\right)^u} = a^x \times \frac{1}{\log_a e} = a^x \times \ln(a)
\end{aligned}$$

$$\boxed{\therefore \frac{d}{dx}[a^x] = a^x \times \ln a}$$

7. **The Logarithmic function** $y = \log_a x$

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{\log_a(x+h)-\log_a x}{h} = \lim_{h\to 0}\frac{\log_a\left(1+\dfrac{h}{x}\right)}{h} = \lim_{h\to 0}\frac{1}{h}\log_a\left(1+\frac{h}{x}\right)$$

Let $u = \dfrac{x}{h} \Rightarrow u \to \infty$ when $h \to 0$ and $h = \dfrac{x}{u}$.

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \lim_{h\to 0}\frac{1}{h}\log_a\left(1+\frac{h}{x}\right) = \lim_{u\to\infty}\frac{u}{x}\log_a\left(1+\frac{1}{u}\right) = \frac{1}{x}\lim_{u\to\infty}\log_a\left(1+\frac{1}{u}\right)^u \\ &= \frac{1}{x}\log_a\left[\lim_{u\to\infty}\left(1+\frac{1}{u}\right)^u\right] = \frac{1}{x}\log_a(e) = \frac{1}{x\ln a}\end{aligned}$$

And we conclude that $\dfrac{d}{dx}[\ln x] = \dfrac{1}{x}$

**Example 2.3.11** *Find the derivative of the function $y = ax + b$ from the first principles.*

*Sol*

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{a(x+h)+b-(ax+b)}{h} = \lim_{h\to 0}\frac{\cancel{ax}+ah+\cancel{b}-\cancel{ax}-\cancel{b}}{h} \\ &= \lim_{h\to 0}\frac{ah}{h} = \lim_{h\to 0} a = a.\end{aligned}$$

**Example 2.3.12** *Find the derivative of the function $y = 4 - 3x$ from the first principles.*



**Example 2.3.13** *Find the derivative of the function $y = x^2 + 2x - 5$ from the first principles.*

*Sol*

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{(x+h)^2+2(x+h)-5-(x^2+2x-5)}{h} \\ &= \lim_{h\to 0}\left[\frac{(x+h)^2-x^2}{h}+\frac{2h}{h}\right] = \lim_{h\to 0}\frac{(x+h)^2-x^2}{h}+\lim_{h\to 0}2 \\ &= \lim_{h\to 0}\frac{h(2x+h)}{h}+\lim_{h\to 0}2 = \lim_{h\to 0}(2x+h)+\lim_{h\to 0}2 = (2x+0)+2 = 2x+2.\end{aligned}$$

**Example 2.3.14** *Find the derivative of the function $y = \dfrac{1}{\sqrt{x}}$ from the first principles.*

*Sol*

$$\begin{aligned}\frac{\Delta y}{h} &= \frac{f(x+h)-f(x)}{h} = \frac{\dfrac{1}{\sqrt{x+h}}-\dfrac{1}{\sqrt{x}}}{h} = \frac{\sqrt{x}-\sqrt{x+h}}{h\left(\sqrt{x}\sqrt{x+h}\right)}\times\frac{\sqrt{x}+\sqrt{x+h}}{\sqrt{x}+\sqrt{x+h}} \\ &= \frac{x-(x+h)}{h\left(\sqrt{x}\sqrt{x+h}\right)\left(\sqrt{x}+\sqrt{x+h}\right)} = \frac{-1}{\left(\sqrt{x}\sqrt{x+h}\right)\left(\sqrt{x}+\sqrt{x+h}\right)} \\ \Rightarrow \frac{dy}{dx} &= \lim_{h\to 0}\frac{-1}{\left(\sqrt{x}\sqrt{x+h}\right)\left(\sqrt{x}+\sqrt{x+h}\right)} = \frac{-1}{\left(\sqrt{x}\sqrt{x}\right)\left(\sqrt{x}+\sqrt{x}\right)} = \frac{-1}{x(2\sqrt{x})} = \frac{-1}{2x^{\frac{3}{2}}}\end{aligned}$$

## 2.4 Properties of Derivatives

1. **Constant multiple rule:** If $f$ is differentiable at $x$ and $c$ is a constant, then

$$\frac{d}{dx}[cf(x)] = cf'(x)$$

2. **Sum Rule:** If $f$ and $g$ are differentiable at $x$, then

$$\frac{d}{dx}[f(x)+g(x)] = f^{'}(x) + g^{'}(x)$$

Proof

Let $F = f + g$, where $f$ and $g$ are differentiable at $x$, so:

$$\begin{aligned}
F^{'}(x) &= \lim_{h\to 0}\frac{F(x+h)-F(x)}{h} = \lim_{h\to 0}\frac{[f(x+h)+g(x+h)]-[f(x)+g(x)]}{h}\\
&= \lim_{h\to 0}\frac{f(x+h)+g(x+h)-f(x)-g(x)}{h} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} + \lim_{h\to 0}\frac{g(x+h)-g(x)}{h}\\
&= f^{'}(x) + g^{'}(x).
\end{aligned}$$

**Note:**

(a) The sum rule can be extended to three or more differentiable functions $f_1, f_2, \dots, f_n$ to obtain the generalized sum rule.

$$\frac{d}{dx}[f_1(x) + f_2(x) + \cdots + f_n(x)] = f_1^{'}(x) + f_2^{'}(x) + \cdots + f_n^{'}(x)$$

(b) The difference of two functions $f - g$ can be rewritten as the sum $f + (-g)$. By combining the sum rule with the constant multiple, the difference rule established.

$$\frac{d}{dx}[f(x)-g(x)] = f^{'}(x) - g^{'}(x)$$

The proof (exercise)

3. **Product Rule:** If $f$ and $g$ are differentiable at $x$, then

$$\frac{d}{dx}[f(x)g(x)] = f(x)g^{'}(x) + f^{'}(x)g(x)$$

Proof

Let $F(x) = f(x).g(x)$, then

$$\begin{aligned}
\frac{dF}{dx} &= \lim_{h\to 0}\frac{F(x+h)-F(x)}{h} = \lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)}{h}\\
&= \lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x+h)+f(x)g(x+h)-f(x)g(x)}{h}\\
&= \lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x+h)}{h} + \lim_{h\to 0}\frac{f(x)g(x+h)-f(x)g(x)}{h}\\
&= \lim_{h\to 0}\left[\underbrace{\frac{f(x+h)-f(x)}{h}}_{f^{'}(x), as\ h\to 0} . \underbrace{g(x+h)}_{g(x), as\ h\to 0}\right] + \lim_{h\to 0}\left[\underbrace{f(x)}_{f(x),\ as\ h\to 0} . \underbrace{\frac{g(x+h)-g(x)}{h}}_{g^{'}(x),\ as\ h\to 0}\right]\\
&= f^{'}(x)g(x) + f(x)g^{'}(x)
\end{aligned}$$

4. **Quotient Rule** Consider the quotient $q(x) = \dfrac{f(x)}{g(x)}$ and note that $f(x) = g(x).q(x)$. By the product rule, we have:

$$f^{'}(x) = g^{'}(x)q(x) + g(x)q^{'}(x)$$

Solving for $q^{'}(x)$, we find that

$$q^{'}(x) = \frac{f^{'}(x) - g^{'}(x)q(x)}{g(x)}$$

Then substituting $q(x)$ with $\dfrac{f(x)}{g(x)}$ we get:

$$q^{'}(x) = \frac{f^{'}(x) - g^{'}(x).\dfrac{f(x)}{g(x)}}{g(x)} = \frac{g(x)f^{'}(x) - g^{'}(x)f(x)}{(g(x))^2}$$

**Theorem 2.4.4** ***Quotient Rule***
*If $f$ and $g$ are differentiable at $x$ , $g(x) \neq 0$, then the derivative of $\frac{f}{g}$ at $x$ exists and*

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f^{'}(x) - g^{'}(x)f(x)}{(g(x))^2}$$

*Proof*

$$\begin{aligned}
\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] &= \lim_{h\to 0}\frac{\dfrac{f(x+h)}{g(x+h)} - \dfrac{f(x)}{g(x)}}{h} = \lim_{h\to 0}\frac{g(x)f(x+h) - f(x)g(x+h)}{h(g(x)g(x+h))} \\
&= \lim_{h\to 0}\frac{g(x)f(x+h) - f(x)g(x+h) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{h(g(x)g(x+h))} \\
&= \frac{\lim\limits_{h\to 0}\dfrac{g(x)[f(x+h) - f(x)]}{h} - \lim\limits_{h\to 0}\dfrac{f(x)[g(x+h) - g(x)]}{h}}{\lim\limits_{h\to 0}(g(x)g(x+h))} \\
&= \frac{g(x)\lim\limits_{h\to 0}\dfrac{[f(x+h) - f(x)]}{h} - f(x)\lim\limits_{h\to 0}\dfrac{[g(x+h) - g(x)]}{h}}{\lim\limits_{h\to 0}(g(x)g(x+h))} \\
&= \frac{g(x)f^{'}(x) - g^{'}(x)f(x)}{(g(x))^2}
\end{aligned}$$

**Note:** $\lim\limits_{h\to 0} g(x+h) = g(x)$ because $g$ is given differentiable and therefore is continuous.

5. **The Chain Rule**

**Theorem 2.4.5** *If $y = f(u)$ is a differentiable function of $u$, and $u = g(x)$ is a differentiable function of $x$, then $y = f(g(x))$ is a differentiable function of $x$ and*

$$\boxed{\frac{d}{dx}[f(g(x))] = f^{'}(g(x)).g^{'}(x)}$$

**Example 2.4.15** *Evaluate $\frac{dy}{dx}$ for the following functions.*

*1.* $y = 4x^3 + \frac{3}{x} + 2\sqrt{x} - 7$

*Sol*

*2.* $y = 3\cos x - \frac{1}{2}\sin x$

*Sol*

*3.* $y = x^7 + \frac{1}{3}x^{-2} - \frac{1}{\sqrt{x}} + \sqrt{x^3}$

*Sol*

*4.* $y = x \sin x$

*Sol*

*5.* $y = x^3 \cos x + x^2 - 5$

*Sol*

*6.* $y = (x + 4)(x^2 - x + 5)$

*Sol*



*7.* $y = \dfrac{3x^2 + 1}{6x + 5}$

*Sol*

*8.* $y = \dfrac{1 + \cos x}{1 - \sin x}$

*Sol*

*9.* $y = 2^{-x} + 3^{x+1}$

*Sol*

*10.* $y = \log_4(x) - 2\ln x$

*Sol*

*11.* $y = x\sin(3x+4)$

*Sol*

*12.* $y = \dfrac{\sin x + \cos x}{\sin x - \cos x}$

*Sol*



*13.* $y = \log_3\left(x + \sqrt{x^2+1}\right)$

*Sol*

**Exercise 2.4.2** *1. From the first principles, evaluate the first derivative for the following function.*

*(i)* $y = x^3$

*(ii)* $y = \dfrac{x}{x^2-5}$

*(iii)* $y = \sin 2x$

*(iv)* $y = xe^{-x}$

2. *Find* $\dfrac{dy}{dx}$ *for the following:*

*(a)* $y = \sin(x \cos 2x)$

*(b)* $y = \dfrac{x^2 + ax + b}{x^2 - ax + b}$



*(c)* $y = \sqrt{1 + \sqrt{2^x - 1}}$

*(d)* $y = \sin^2 \left( \dfrac{2x}{x+1} \right)$

*(e)* $y = \dfrac{1 + 2\tan^2 x}{1 - \tan^2 x}$

(f) $y = \ln(\sec x + \tan x)$

(g) $y = \dfrac{1}{\sqrt{1 + \sqrt{x}}}$

(h) $y = \dfrac{x^2}{\sqrt{a^2 - x^2}}$

(i) $y = xe^{-x^2}$

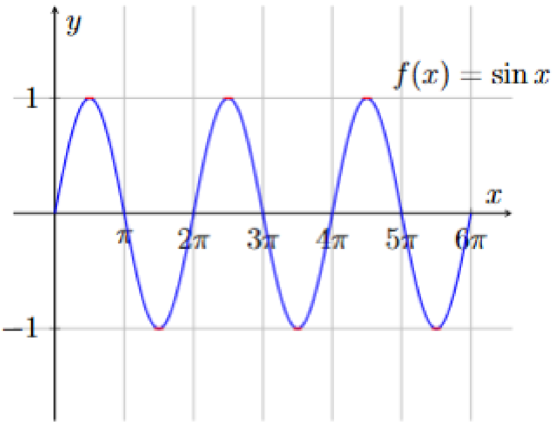


(j) $y = x\cos(\sin 2x)$

# Lecture Four

# 1 Trigonometric Functions

## 1.1 Sine Function

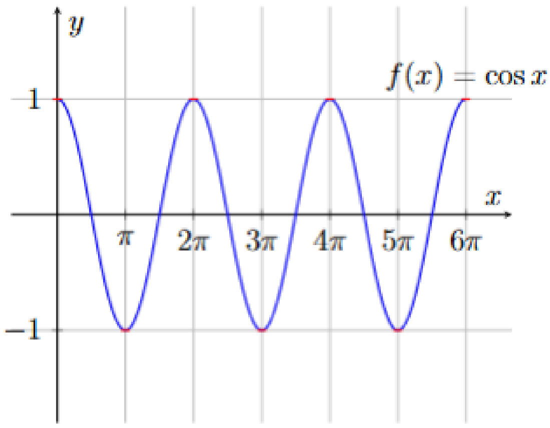



**Properties of $y = \sin(x)$:**

- $\forall x \in \mathbb{R},\ \sin(-x) = -\sin(x)$
- $\forall x \in \mathbb{R},\ \sin(x + 2\pi) = \sin(x)$
- $\forall k \in \mathbb{Z},\ \sin(x + 2k\pi) = \sin(x)$
- $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$
- $\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$

## 1.2 Cosine Function

$y = \cos x$ is the cosine function, and we know the domain $D = \mathbb{R}$ and the range $R = [-1, 1]$ and it is an even function.

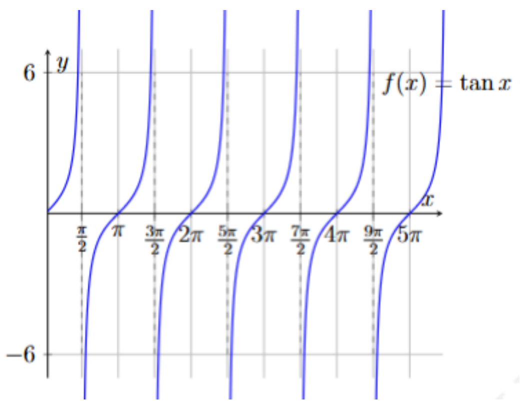



**Properties of $y = \cos(x)$:**

- $\forall x \in \mathbb{R},\ \cos(-x) = \cos x$
- $\forall x \in \mathbb{R},\ \cos(x + 2\pi) = \cos x$
- $\forall k \in I,\ \cos(x + 2k\pi) = \cos x$
- $\forall x \in \mathbb{R},\ \cos\left(\frac{\pi}{2} - x\right) = \sin x$
- $\forall x \in \mathbb{R},\ \cos\left(\frac{\pi}{2} + x\right) = -\sin x$

## 1.3 Tangent Function

$y = \tan x = \dfrac{\sin x}{\cos x}$ is the tangent function, we know the domain $D = \mathbb{R} - \left\{\dfrac{(2k-1)\pi}{2}, k \in I\right\}$ and the range $R = \mathbb{R}$ and it is an odd function.

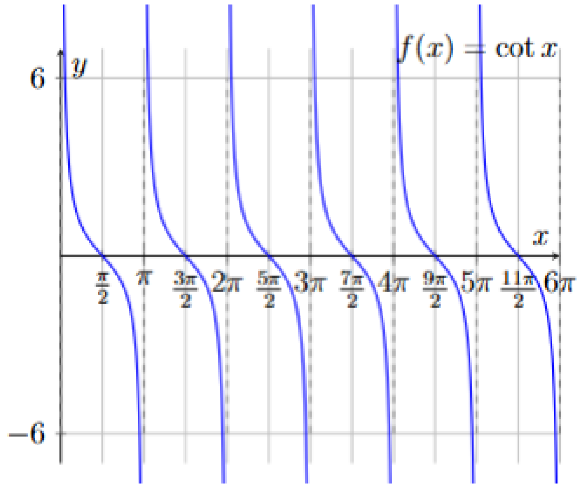



**Properties of $y = \tan(x)$:**

- $\forall x \in D,\ \tan(-x) = -\tan x$
- $\forall x \in D,\ \tan(x + 2\pi) = \tan x$
- $\forall k \in I, x \in D,\ \tan(x + 2k\pi) = \tan x$
- $\forall x \in D,\ \tan\left(\frac{\pi}{2} - x\right) = \cot x$
- $\forall x \in D,\ \cos\left(\frac{\pi}{2} + x\right) = -\cot x$

## 1.4 Cotangent Function

$y = \cot x = \dfrac{\cos x}{\sin x}$ is the cotangent function, and we know the domain $D = \mathbb{R} - \{k\pi, k \in I\}$ and the range $R = \mathbb{R}$ and it is an odd function.

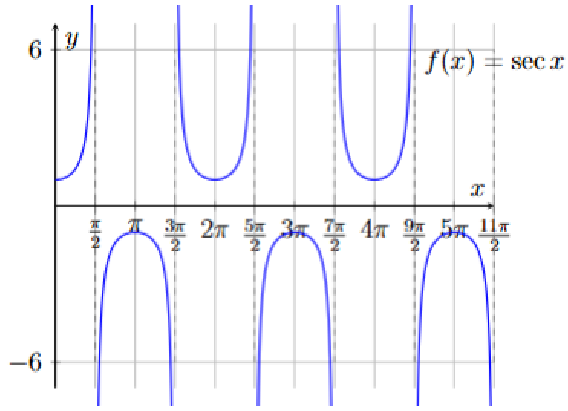



**Properties of $y = \cot(x)$:**

- $\forall x \in D,\ \cot(-x) = -\cot x$
- $\forall x \in D,\ \cot(x + 2\pi) = \cot x$
- $\forall k \in I, x \in D,\ \cot(x + 2k\pi) = \cot x$
- $\forall x \in D,\ \cot\left(\dfrac{\pi}{2} - x\right) = \tan x$
- $\forall x \in D,\ \cot\left(\dfrac{\pi}{2} + x\right) = -\tan x$

## 1.5 Secant Function

$y = \sec x = \dfrac{1}{\cos x}$ is the secant function, and we know the domain $D = \mathbb{R} - \left\{\dfrac{(2k-1)\pi}{2}, k \in I\right\}$ and the range $R = \mathbb{R} - (-1, 1)$ and it is an even function.

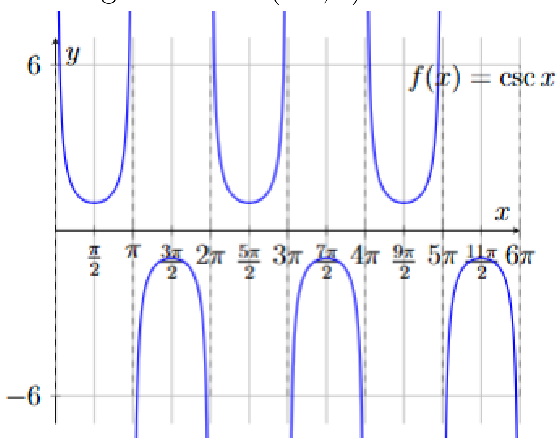



**Properties of $y = \sec(x)$:**

- $\forall x \in D,\ \sec(-x) = \sec x$
- $\forall x \in D,\ \sec(x + 2\pi) = \sec x$
- $\forall k \in I, x \in D,\ \sec(x + 2k\pi) = \sec x$
- $\forall x \in D,\ \sec\left(\dfrac{\pi}{2} - x\right) = \csc x$
- $\forall x \in D,\ \sec\left(\dfrac{\pi}{2} + x\right) = -\csc x$

## 1.6 Cosecant Function

$y = \csc x = \dfrac{1}{\sin x}$ is the cosecant function, we know the domain $D = \mathbb{R} - \left\{\dfrac{(2k-1)\pi}{2}, k \in I\right\}$ and the range $R = \mathbb{R} - (-1, 1)$ and it is an odd function.




**Properties of $y = \csc(x)$:**

- $\forall x \in D,\ \csc(-x) = -\csc x$
- $\forall x \in D,\ \csc(x + 2\pi) = \csc x$
- $\forall k \in I, x \in D,\ \sec(x + 2k\pi) = \csc x$
- $\forall x \in D,\ \csc\left(\dfrac{\pi}{2} - x\right) = \sec x$
- $\forall x \in D,\ \csc\left(\dfrac{\pi}{2} + x\right) = \sec x$

# 2 Differentiation of Trigonometric Functions

## 2.1 The Rules

1. $\frac{d}{dx}[\sin x] = \cos x$

2. $\frac{d}{dx}[\cos x] = -\sin x$

3. $\frac{d}{dx}[\tan x] = \sec^2 x$

Proof

$$\frac{d}{dx}[\tan x] = \frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] = \frac{\cos x \times \cos x - \sin x \times -\sin x}{\cos^2 x}$$
$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

4. $\frac{d}{dx}[\cot x] = -\csc^2 x$

Proof

$$\frac{d}{dx}[\cot x] = \frac{d}{dx}\left[\frac{\cos x}{\sin x}\right] = \frac{\sin x \times -\sin x - \cos x \times \cos x}{\sin^2 x} = -\frac{\sin^2 x + \cos^2 x}{\sin^2 x}$$
$$= -\frac{1}{\sin^2 x} = -\csc^2 x$$

5. $\frac{d}{dx}[\sec x] = \sec x . \tan x$

Proof

$$\frac{d}{dx}[\sec x] = \frac{d}{dx}\left[\frac{1}{\cos x}\right] = \frac{\cos x \times 0 - 1 \times -\sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x}$$
$$= \frac{1}{\cos x} . \frac{\sin x}{\cos x} = \sec x . \tan x$$

6. $\frac{d}{dx}[\csc x] = -\csc x . \cot x$

Proof

$$\frac{d}{dx}[\csc x] = \frac{d}{dx}\left[\frac{1}{\sin x}\right] = \frac{\sin x \times 0 - 1 \times \cos x}{\sin^2 x} = -\frac{\cos x}{\sin^2 x}$$
$$= -\frac{1}{\sin x} . \frac{\cos x}{\sin x} = -\csc x . \cot x$$

**In general:**

| No. | The Trig. Function | Derivative |
|---|---|---|
| (1) | $\frac{d}{dx}[\sin u(x)]$ | $u^{'}(x)\cos u(x)$ |
| (2) | $\frac{d}{dx}[\cos u(x)]$ | $-u^{'}(x).\sin u(x)$ |
| (3) | $\frac{d}{dx}[\tan u(x)]$ | $u^{'}(x)\sec^2 u(x)$ |
| (4) | $\frac{d}{dx}[\cot u(x)]$ | $-u^{'}(x)\csc^2 u(x)$ |
| (5) | $\frac{d}{dx}[\sec u(x)]$ | $u^{'}(x)\sec u(x)\tan u(x)$ |
| (6) | $\frac{d}{dx}[\csc u(x)]$ | $-u^{'}(x)\csc u(x)\cot u(x)$ |

## 2.2 Examples

**Example 2.2.1** *Find* $\frac{dy}{dx}$ *for the following functions:*

1. $y = 3\sin(2x^2+5) - \cos^3 x$

*Sol*

$$\frac{dy}{dx} = 3\times 4x \times \cos(2x^2+5) - 3\times\cos^2 x(-\sin x) = 12x\cos(2x^2+5) + 3\sin x\cos^2 x$$

2. $y = 4\sec(\cos x^2 + 1)$

*Sol*

$$\begin{aligned}\frac{dy}{dx} &= 4\left(-2x\sin x^2\right)\sec(\cos x^2+1)\tan(\cos x^2+1) \\ &= -8x\sin x^2\sec(\cos x^2+1)\tan(\cos x^2+1)\end{aligned}$$

3. $y = \sqrt{\tan(x\sec x)}$

*Sol*

*Let* $w = \tan(x\sec x) \Rightarrow y = \sqrt{w}$ *i.e.* $y = f(w)$. *And let* $u = x\sec x$ *i.e.* $u = h(x) \Rightarrow w = \tan u$ *i.e.* $w = g(u)$. *Hence*

$$\frac{dy}{dx} = \frac{dy}{dw}.\frac{dw}{du}.\frac{du}{dx}$$

*Now:* $\frac{dy}{dw} = \frac{1}{2\sqrt{w}}$ , $\frac{dw}{du} = \sec^2 u$ , $\frac{du}{dx} = x\sec x\tan x + \sec x$. *Therefore:*

$$\frac{dy}{dx} = \frac{1}{2\sqrt{w}} \times \sec^2 u \times (x\sec x\tan x + \sec x) = \frac{\sec^2(x\sec x)}{2\sqrt{\tan(x\sec x)}}.(x\sec x\tan x + \sec x)$$

4. $y = \left(\frac{\sin x}{\cot x + \tan x}\right)$

*Sol*

*We can write* $\dfrac{\sin x}{\cot x + \tan x}$ *as:*

$$\frac{\sin x}{\cot x + \tan x} = \frac{\sin x}{\dfrac{\cos x}{\sin x} + \dfrac{\sin x}{\cos x}} = \frac{\sin x}{\dfrac{\cos^2 x + \sin^2 x}{\sin x \cos x}} = \sin^2 x \cos x$$

*Now* $y = \csc(\sin^2 x \cos x)$, *let* $u = \sin^2 x \cos x \Rightarrow y = \csc u$ *and* $\dfrac{dy}{dx} = \dfrac{dy}{du} . \dfrac{du}{dx}$. *Therefore:*

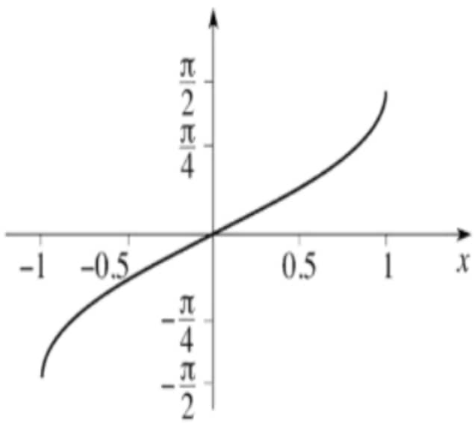
$$\frac{dy}{du} = -\csc u \cot u \quad , \frac{du}{dx} = -\sin^3 x + \cos x(2\sin x \cos x) = \sin 2x \cos x - \sin^3 x$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= -\csc u \cot u \times (\sin 2x \cos x - \sin^3 x) \\ &= -(\sin 2x \cos x - \sin^3 x)\csc(\sin^2 x \cos x)\cot(\sin^2 x \cos x) \end{aligned}$$

# 3 The Inverse Trigonometric Functions

## 3.1 The Inverse Sine Function

Let $x = \sin y$, then the inverse sine function is $y = \arcsin(x) = \sin^{-1} x$, the domain of definition is $D = [-1, 1]$, the range $R = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and it is an odd function, where $\sin^{-1}(-x) = -\sin^{-1} x$.




### The derivative of inverse sin function

$$y = \sin^{-1} x \Rightarrow x = \sin y \Rightarrow 1 = \cos y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$

And we know that $\cos y = \pm\sqrt{1 - \sin^2 y}$. Since $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \cos y > 0$, hence

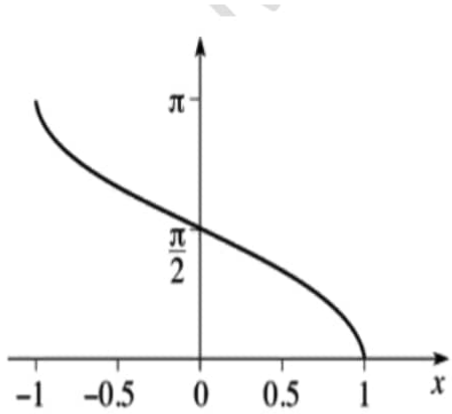
$$\cos y = \sqrt{1 - \sin^2 y}$$

Therefore

$$\boxed{\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}}$$

## 3.2 The Inverse Cosine Function

Let $x = \cos y$, then the inverse cosine function is $y = \arccos(x) = \cos^{-1} x$, the domain of definition is $D = [-1, 1]$, the range $R = [0, \pi]$, and it is an odd function, where $\cos^{-1}(-x) = \cos^{-1} x$ .




### The derivative of inverse cos function

$$y = \cos^{-1} x \Rightarrow x = \cos y \Rightarrow 1 = -\sin y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{-1}{\sin y}$$

And we know that $\sin y = \pm\sqrt{1 - \cos^2 y}$. Since $y \in [0, \pi] \Rightarrow \sin y > 0$, hence

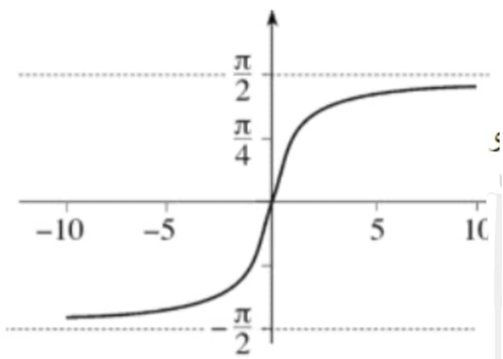
$$\sin y = \sqrt{1 - \cos^2 y}$$

Therefore

$$\boxed{\frac{dy}{dx} = \frac{-1}{\sqrt{1 - x^2}}}$$

## 3.3 The Inverse Tangent Function

Let $x = \tan y$, then the inverse tangent function is $y = \arctan(x) = \tan^{-1} x$, the domain of definition is $D = \mathbb{R}$, the range $R = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, and it is an odd function, where $\tan^{-1}(-x) = -\tan^{-1} x$.




### The derivative of inverse tan function

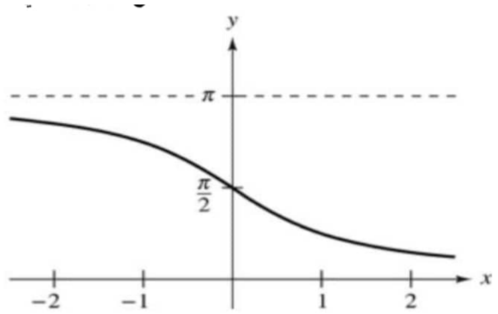
$$y = \tan^{-1} x \Rightarrow x = \tan y \Rightarrow 1 = \sec^2 y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

We know $\sec^2 y = 1 + \tan^2 y = 1 + x^2$

$$\therefore \boxed{\frac{dy}{dx} = \frac{1}{1 + x^2}}$$

## 3.4 The Inverse Cotangent Function

Let $x = \cot y$, then the inverse tangent function is $y = \cot^{-1} x$, the domain of definition is $D = \mathbb{R}$, the range $R = (0, \pi)$, and it is an odd function, where $\cot^{-1}(-x) = -\cot^{-1} x$.




### The derivative of inverse cot function

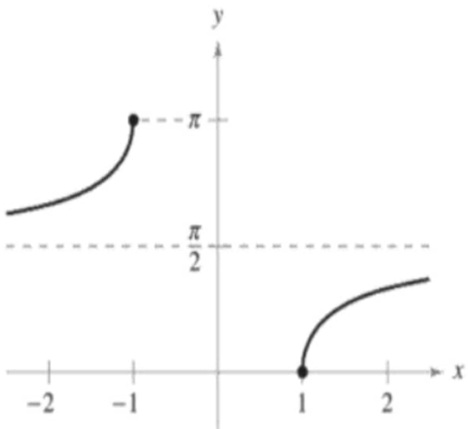
$$y = \cot^{-1} x \Rightarrow x = \cot y \Rightarrow 1 = -\csc^2 y\,\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{-1}{\csc^2 y}$$

We know $\csc^2 y = 1 + \cot^2 y = 1 + x^2$

$$\therefore \boxed{\frac{dy}{dx} = \frac{-1}{1 + x^2}}$$

## 3.5 The Inverse Secant Function

Let $x = \sec y$, then the inverse cosine function is $y = \sec^{-1} x$, the domain of definition is $D = \mathbb{R} - (-1, 1)$, the range $R = \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$, and it is an even function, where $\sec^{-1}(-x) = \sec^{-1} x$ .




### The derivative of inverse sec function

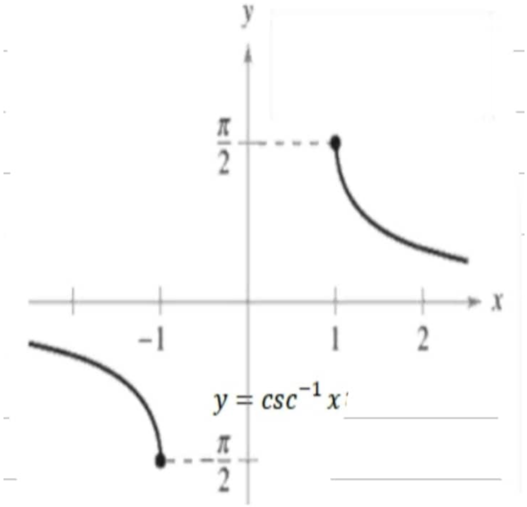
$$y = \sec^{-1} x \Rightarrow x = \sec y \Rightarrow 1 = \sec y\ \tan y\,\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\sec y\ \tan y}$$

And we know $\tan y = \pm\sqrt{\sec^2 y - 1}$, either $y \in \left[0, \frac{\pi}{2}\right)$ this implies that $\tan y > 0$ and we take $\tan y = \sqrt{\sec^2 y - 1}$ and thus $\sec y\ \tan y = |\sec y|\sqrt{\sec^2 y - 1} > 0$ or $y \in \left(\frac{\pi}{2}, \pi\right]$, which is implies that $\tan y$ & $\sec y < 0 \Rightarrow \tan y = -\sqrt{\sec^2 y - 1}$ and thus $\sec y \tan y = |\sec y|\sqrt{\sec^2 y - 1} > 0$. Now

$$\frac{dy}{dx} = \frac{1}{|\sec y|\sqrt{\sec^2 y - 1}} \Rightarrow \boxed{\frac{dy}{dx} = \frac{1}{|x|\sqrt{x^2 - 1}}}$$

## 3.6 The Inverse Cosecant Function

Let $x = \csc y$, then the inverse cosine function is $y = \csc^{-1} x$, the domain of definition is $D = \mathbb{R} - (-1, 1)$, the range $R = \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$, and it is an odd function, where $\csc^{-1}(-x) = -\csc^{-1} x$ .




## The derivative of inverse csc function

$$y = \csc^{-1} x \Rightarrow x = \csc y \Rightarrow 1 = -\csc y \ \cot y \ \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{-1}{\csc y \ \cot y}$$

And we know $\cot y = \pm\sqrt{\csc^2 y - 1}$, either $y \in \left[-\frac{\pi}{2}, 0\right)$ this implies that $\csc y > 0$ and we take $\cot y = -\sqrt{\sec^2 y - 1}$ and thus $\csc y \ \cot y = |\csc y|\sqrt{\csc^2 y - 1} > 0$ or $y \in \left(0, \frac{\pi}{2}\right]$, which is implies that $\csc y > 0 \Rightarrow \cot y = \sqrt{\csc^2 y - 1}$ and thus $\csc y \cot y = |\csc y|\sqrt{\csc^2 y - 1} > 0$. Now

$$\frac{dy}{dx} = \frac{-1}{|\csc y|\sqrt{\csc^2 y - 1}} \Rightarrow \boxed{\frac{dy}{dx} = \frac{-1}{|x|\sqrt{x^2 - 1}}}$$

**Example 3.6.2** *Find $\frac{dy}{dx}$ for the following functions:*

1. $y = \frac{x}{2} \sin^{-1} 2x$

*Sol*

$$\frac{dy}{dx} = \frac{x}{2} . \frac{2}{\sqrt{1 - 4x^2}} + \frac{1}{2} \sin^{-1} 2x = \frac{x}{\sqrt{1 - 4x^2}} + \frac{1}{2} \sin^{-1} 2x$$

2. $y = \cos^{-1}(x \cos x)$

*Sol*

*Let $u = x \cos x \Rightarrow u'(x) = \cos x - x \sin x$ and we know $\frac{d}{dx}\left(\cos^{-1}(u(x))\right) = \frac{-u'(x)}{\sqrt{1 - (u(x))^2}}$*

$$\therefore \frac{dy}{dx} = \frac{-(\cos x - x \sin x)}{\sqrt{1 - (x \cos(x))^2}} = \frac{x \sin x - \cos x}{\sqrt{1 - x^2 \cos^2 x}} \qquad \square$$

3. $y = \frac{\tan^{-1} x}{x^2 + 1}$

*Sol*

$$\frac{dy}{dx}=\left(\frac{(x^2+1)\times\dfrac{1}{x^2+1}-\tan^{-1}x\times(2x)}{(x^2+1)^2}\right)=\frac{1-2x\tan^{-1}x}{(x^2+1)^2}$$

*4.* $y=\tan^{-1}\left(\dfrac{\tan x+1}{\tan x-1}\right)$

*Sol*

*We know that* $\dfrac{d}{dx}\left[\tan^{-1}(u(x))\right]=\dfrac{u'(x)}{1+(u(x))^2}$

$$\begin{aligned}\therefore\frac{dy}{dx}&=\frac{1}{1+\left(\dfrac{\tan x+1}{\tan x-1}\right)^2}\times\frac{(\tan x-1)\times\sec^2 x-(\tan x+1)\times\sec^2 x}{(\tan x-1)^2}\\&=\left(\frac{(\tan x-1)^2}{(\tan x+1)^2+(\tan x-1)^2}\right)\times\frac{\sec^2 x[\tan x-1-\tan x-1]}{(\tan x-1)^2}\\&=\frac{-2\sec^2 x}{2(\tan^2 x+1)}=\frac{-2\sec^2 x}{2\sec^2 x}=-1\end{aligned}$$

***Another Solution:*** *We can write* $\dfrac{\tan x+1}{\tan x-1}$ *as:*

$$\begin{aligned}\frac{\tan x+1}{\tan x-1}&=\frac{\dfrac{\sin x}{\cos x}+1}{\dfrac{\sin x}{\cos x}-1}=\frac{\sin x+\cos x}{\sin x-\cos x}=\frac{\sin^2 x+2x\sin x\cos x+\cos^2 x}{\sin^2 x-\cos^2 x}\\&=-\left[\frac{1+\sin 2x}{\cos 2x}\right]=-[\sec 2x+\tan 2x]\end{aligned}$$

*Now,* $\tan^{-1}\left(\dfrac{\tan x+1}{\tan x-1}\right)=-\tan^{-1}[\sec 2x+\tan 2x]$

$$\begin{aligned}\therefore\frac{dy}{dx}&=-\left(\frac{2\sec 2x\tan 2x+2\sec^2 x}{1+(\sec 2x+\tan 2x)^2}\right)=-\left(\frac{2\sec 2x\tan 2x+2\sec^2 x}{1+\sec^2 x+2\sec 2x\tan 2x+\tan^2 x}\right)\\&=-\left(\frac{2\sec 2x\tan 2x+2\sec^2 x}{2\sec 2x\tan 2x+2\sec^2 x}\right)=-1.\end{aligned}$$

***Another Solution:***

$$\begin{aligned}y=&\tan^{-1}\left(\frac{\tan x+1}{\tan x-1}\right)=-\tan^{-1}\left(\frac{\tan x+1}{1-\tan x}\right)=-\tan^{-1}\left(\frac{\tan x+\tan\frac{\pi}{4}}{1-\tan x\times\tan\frac{\pi}{4}}\right)\\=&-\tan^{-1}\left(\tan\left(x+\frac{\pi}{4}\right)\right)=-\left(x+\frac{\pi}{4}\right)\Rightarrow\boxed{y=-x-\frac{\pi}{4}\Rightarrow\frac{dy}{dx}=-1}\end{aligned}$$

**Exercise 3.6.1** *Find* $\dfrac{dy}{dx}$ *for the following functions.*

*1.*

$$y=2\sin^{-1}(x-1)$$

*Sol*

2.

$$y = \tan^{-1}\left(\frac{x}{a}\right) + 2\sin^{-1}\left(\frac{a}{x}\right)$$

*Sol*

3.

$$y = \frac{\cos^{-1} x^2}{x+1}$$

*Sol*



4.

$$y = e^{x^2} \sin^{-1} x$$

*Sol*

5.

$$y = \sin^{-1} 2x + \cos^{-1} 2x$$

*Sol*

6.

$$y = x\cos^{-1} x - \sqrt{1-x^2}$$

*Sol*

*7.*

$$y = \frac{1}{2}\left[x\sqrt{4-x^2} + 4\sin^{-1}\left(\frac{x}{2}\right)\right]$$

*Sol*

*8.*

$$y = \cos x . \sec^{-1} x$$

*Sol*



*9.*

$$y = e^{\sin^{-1} x} + \frac{1}{\sqrt{1+x^2}}$$

*Sol*

*10.*

$$y = \csc\left[\frac{x}{x+1}\right] - \sec^{-1}\left[x + \frac{1}{x}\right]$$

*Sol*

*11.*

$$y = \frac{\cot^{-1} x}{1 + x^2}$$

*Sol*

*12.*

$$y = \ln (x - \tan x)$$

*Sol*

*13.*

$$y = \csc^{-1} \left[ \frac{1 - x}{x + 1} \right]$$

*Sol*



*14.*

$$y = \ln \sqrt{\frac{x + 1}{x - 1}} + \tan^{-1} 2x$$

*Sol*

# Lecture Five

# 1 Exponential and Logarithmic Functions

## 1.1 Exponential Function

**Definition 1.1.1** *Let $a > 0,\ a \neq 1$, then the exponential function of the base $a$ ia $y = a^x$.*

### 1.1.1 Some Properties of the exponential function

For $y = a^x$,

1. If $a > 1$, then:

   (a) $a^x > 0\ , \forall x \in \mathbb{R}$

   (b) $y = a^x$ is an increasing function.

   (c) $\lim\limits_{x \to -\infty} a^x \to 0$

   (d) $\lim\limits_{x \to +\infty} a^x \to \infty$

   (e) $f(0) = a^0 = 1,\ \forall a > 1$, the graph of $a^x$ passes through the point (0,1) on $y-$axis.

2. If $0 < a < 1$, then:

   (a) $a^x > 0\ , \forall x \in \mathbb{R}$

   (b) $y = a^x$ will be decreasing function.

   (c) $\lim\limits_{x \to -\infty} a^x \to \infty$

   (d) $\lim\limits_{x \to +\infty} a^x \to 0$

   (e) $f(0) = a^0 = 1,\ \forall a < 1$, the graph of $a^x$ passes through the point (0,1) on $y-$axis.



**Theorem 1.1.1** *Prove that*

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^k}{k!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

*Proof*

*We know* $e^x = \lim\limits_{n \to \infty} \left(1 + \frac{x}{n}\right)^n$, *now:*

$$\begin{aligned}
\left(1 + \frac{x}{n}\right)^n &= 1 + \binom{n}{1}\frac{x}{n} + \binom{n}{2}\frac{x^2}{n^2} + \binom{n}{3}\frac{x^3}{n^3} + \cdots + \binom{n}{k}\frac{x^k}{n^k} + \ldots \\
&= 1 + n.\frac{x}{n} + \frac{n(n-1)}{2!}\frac{x^2}{n^2} + \frac{n(n-1)(n-2)}{3!}\frac{x^3}{n^3} + \cdots + \frac{n(n-1)(n-2)\ldots}{k!}\frac{x^k}{n^k} + \ldots \\
&=1 + x + \frac{1}{2!}\left(1 - \frac{1}{n}\right)x^2 + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)x^3 + \cdots + \\
&\quad \frac{1}{k!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\ldots\left(1 - \frac{k-1}{n}\right)x^k
\end{aligned}$$

*Therefore* $\left(1+\dfrac{x}{n}\right)^n = \displaystyle\sum_{k=0}^{\infty} \frac{x^k}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{k-1}{n}\right)$ *and this implies that:*

$$\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = \sum_{k=0}^{\infty} \lim_{n\to\infty} \frac{x^k}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{k-1}{n}\right) =$$

$$\boxed{\Rightarrow e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = \sum_{k=0}^{\infty}\frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots}$$

**Example 1.1.1** *Evaluate the following sums using the definition of exponential function $e^x$:*

1. $\displaystyle\sum_{k=1}^{\infty}\frac{2^k}{k!}$

*Sol*

$$\sum_{k=1}^{\infty}\frac{2^k}{k!} = \left(1+\sum_{k=1}^{\infty}\frac{2^k}{k!}\right) - 1 = \frac{2^0}{0!} + \sum_{k=1}^{\infty}\frac{2^k}{k!} - 1 = \sum_{k=0}^{\infty}\frac{2^k}{k!} - 1 = e^2 - 1.$$

2. $\displaystyle\sum_{k=2}^{\infty}\frac{(-1)^k}{k!}$



*Sol*

$$\sum_{k=2}^{\infty}\frac{(-1)^k}{k!} = 1 - 1 + \sum_{k=2}^{\infty}\frac{(-1)^k}{k!} = \frac{(-1)^0}{0!} + \frac{(-1)^1}{1!} + \sum_{k=2}^{\infty}\frac{(-1)^k}{k!} = \sum_{k=0}^{\infty}\frac{(-1)^k}{k!} = e^{-1}.$$

3. $\displaystyle\sum_{k=2}^{\infty}\frac{(-1)^k}{2^{k-1}k!}$

*Sol*

$$\sum_{k=2}^{\infty}\frac{(-1)^k}{2^{k-1}k!} = \sum_{k=2}^{\infty} 2\times\frac{(-1)^k}{4^k k!} = 2\times\sum_{k=2}^{\infty}\frac{\left(\frac{-1}{4}\right)^k}{k!} = 2\times\left[\sum_{k=2}^{\infty}\frac{\left(\frac{-1}{4}\right)^k}{k!} - 1 + \frac{1}{4}\right]$$

$$= 2\left(e^{-\frac{1}{4}} - \frac{3}{4}\right) = \frac{2}{\sqrt[4]{e}} - \frac{3}{2}.$$

**Example 1.1.2** *Find the following limits using $e^x$ definition.*

1. $\displaystyle\lim_{x\to 0}\frac{1-e^{2x}+2x}{x^2}$

*Sol*

*We know that* $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$

*This implies that* $e^{2x} = 1 + 2x + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \cdots = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \ldots$. *Now:*

$$\frac{1 - e^{2x} + 2x}{x^2} = \frac{1 + 2x - \left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \ldots\right)}{x^2} = -\frac{\left(2x^2 + \frac{4}{3}x^3 + \ldots\right)}{x^2}$$
$$= -\left(2 + \frac{4}{3}x + \ldots\right)$$

$$\therefore \lim_{x\to 0} \frac{1 - e^{2x} + 2x}{x^2} = -\lim_{x\to 0}\left(2 + \frac{4}{3}x + \ldots\right) = -2.$$

*2.* $\lim_{x\to 0} \frac{e^{ax} - 1}{\sin bx} \quad , b \neq 0$

*Sol*

*We can write the numerator as follows:*

$$e^{ax} - 1 = \sum_{k=0}^{\infty} \frac{(ax)^k}{k!} - 1 = \left(1 + ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \ldots\right) - 1 = ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \ldots$$

$$\therefore \lim_{x\to 0} \frac{e^{ax} - 1}{\sin bx} = \lim_{x\to 0} \frac{ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \ldots}{\sin bx} = \lim_{x\to 0} \frac{\frac{ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \ldots}{x}}{\frac{\sin bx}{x}}$$

$$= \frac{\lim_{x\to 0} \frac{ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \ldots}{x}}{\lim_{x\to 0} \frac{\sin bx}{x}} = \frac{a}{b}$$

*3.* $\lim_{x\to 0} \frac{a^x - b^x}{x}$

*Sol*

*For you*

### 1.1.2 Derivative of the Exponential Function

1.
$$\boxed{\frac{d}{dx}\left[a^{u(x)}\right] = a^{u(x)} \times \ln a \times \frac{du}{dx}}$$

2.
$$\boxed{\frac{d}{dx}\left[e^{u(x)}\right] = e^{u(x)} \times \ln e \times \frac{du}{dx} = u^{'}(x)e^{u(x)}}$$

And when $u(x) = x \Rightarrow$

1.
$$\frac{d}{dx}[a^x] = a^x \ln a$$

2.
$$\frac{d}{dx}[e^x] = e^x$$

## 1.2 Logarithmic Functions

**Definition 1.2.2** *Let $a > 0$, $a \neq 1$. The **logarithmic function** of the base $a$ is $y = \log_a x$, and it is the inverse of $y = a^x$. The domain of definition of it $D = \mathbb{R}^+ = (0, \infty)$ and $R = \mathbb{R}$.*

**Note** that if the base is $e$, it is called the natural logarithm, and we denoted by $\ln x$ ($\log_e x = \ln x$) and thus $\ln x$ is the inverse of $e^x$.

### 1.2.1 Properties:

| No. | log | ln |
|---|---|---|
| (1) | $\log_a 1 = 0$ | $\ln 1 = 0$ |
| (2) | $\log_a x + \log_a y = \log_a xy$ | $\ln x + \ln y = \ln xy$ |
| (3) | $\log_a x^n = n \log_a x$ | $\ln x^n = n \ln x$ |
| (4) | $\log_a \frac{1}{x} = -\log_a x$ | $\ln \frac{1}{x} = -\ln x$ |
| (5) | $\log_a x - \log_a y = \log_a \frac{x}{y}$ | $\ln x - \ln y = \ln \frac{x}{y}$ |
| (6) | $\log_a b = \frac{1}{\log_b a}$ | $\ln b = \frac{1}{\log_b e}$ |
| (7) | $\log_{a^n} x = \frac{1}{n} \log_a x$ | $\log_{e^n} x = \frac{1}{n} \ln x$ |
| (8) | $\log_{a^n} x^n = \log_a x$ | $\log_{e^n} x^n = \ln x$ |



### 1.2.2 Derivative of Logarithmic Function

1. $$\boxed{\frac{d}{dx}[\log_a u(x)] = \frac{1}{u(x)\ln a} \times \frac{du}{dx}}$$

2. $$\boxed{\frac{d}{dx}[\ln u(x)] = \frac{1}{u(x)} \times \frac{du}{dx}}$$

And if $u(x) = x \Rightarrow$

1. $$\boxed{\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}}$$

2. $$\boxed{\frac{d}{dx}[\ln x] = \frac{1}{x}}$$

**Derivative of** $[f(x)]^{g(x)}$:

Let $y = (f(x))^{g(x)}$, then the derivative evaluated as follows:

$$y = [f(x)]^{g(x)} \Rightarrow \ln y = g(x) \ln f(x)$$

Now, by differentiating both sides w.r.t $x$, we get:

$$\frac{y^{'}}{y} = g(x) \times \frac{f^{'}(x)}{f(x)} + g^{'}(x)\ln f(x)$$

Therefore the derivative will be:

$$\boxed{y^{'} = \left[g(x) \times \frac{f^{'}(x)}{f(x)} + g^{'}(x)\ln f(x)\right] \times y}$$

**Example 1.2.3** *Find the first derivative of the following function:*

1. $y = e^{\sin^2(4x)}$

*Sol*

$$\frac{dy}{dx} = 2(\sin 4x) \times (4\cos 4x) \times e^{\sin^2(4x)} = 8\sin 4x \cos 4x \; e^{\sin^2(4x)}.$$

2. $y = e^{\frac{2x^2+8}{4x+1}}$

*Sol*

$$\frac{dy}{dx} = \left(\frac{(4x+1)(4x) - (2x^2+8)(4)}{(4x+1)^2}\right) e^{\frac{2x^2+8}{4x+1}} = \frac{4x - 8x^2 - 12}{(4x+1)^2} e^{\frac{2x^2+8}{4x+1}}.$$

3. $y = x^3 e^{2x^2-1}$

*Sol*

$$\frac{dy}{dx} = x^3\left(4x \times e^{2x^2-1}\right) + e^{2x^2-1} \times (3x^2) = 4x^4 e^{2x^2-1} + 3x^2 e^{2x^2-1}$$

4. $y = e^{\frac{\tan x}{\sin(\cos x)}}$

*Sol*

$$\frac{dy}{dx} = \left[\frac{\sin(\cos x)\sec^2 x + \tan x(\sin x \cos(\cos x))}{\sin^2(\cos x)}\right] e^{\frac{\tan x}{\sin(\cos x)}}$$

**Example 1.2.4** *Find* $\frac{dy}{dx}$ *for the following functions:*

(a) $y = 2^x + \frac{1}{\sqrt{3^x}}$

*Sol*

$$y = 2^x + 3^{-\frac{x}{2}} \Rightarrow \frac{dy}{dx} = 2^x \ln 2 + 3^{-\frac{x}{2}} \times \ln 3 \times \left(-\frac{1}{2}\right) = 2^x \ln 2 - \frac{\ln 3}{2\sqrt{3^x}}$$

(b) $y = \frac{4^x - 1}{4^x + 1}$

*Sol*

$$\begin{aligned}\frac{dy}{dx} &= \frac{(4^x+1)(4^x\ln 4)-(4^x-1)(4^x\ln 4)}{(4^x+1)^2}\\ &= \frac{4^x\ln 4(4^x+1-4^x+1)}{(4^x+1)^2} = \frac{2^{2x+1}\ln 4}{(4^x+1)^2}\end{aligned}$$

*(c)* $y = x^{\sec^{-1}x}$

*Sol*

*By tanking* ln *for both sides* $\Rightarrow$ $\ln y = \sec^{-1}x\ln x$, *now:*

$$\frac{1}{y}\frac{dy}{dx} = \frac{sec^{-1}x}{x} + \frac{\ln x}{|x|\sqrt{x^2-1}}$$

*Therefore, the required derivative is:*

$$\frac{dy}{dx} = \left[\frac{sec^{-1}x}{x} + \frac{\ln x}{|x|\sqrt{x^2-1}}\right]x^{\sec^{-1}x}$$

*(d)* $y = \dfrac{(x+1)^3(x^2+4)^8}{\sqrt[4]{x^2+2}}$

*Sol*

*Firstly we take the* ln *for both sides:*

$$\begin{aligned}\ln y = \; &\ln\left[\frac{(x+1)^3(x^2+4)^8}{\sqrt[4]{x^2+2}}\right]\\ &= 3\ln(x+1) + 8\ln(x^2+4) - \frac{1}{4}\ln(x^2+2)\end{aligned}$$

*Then we differentiate both sides, we get:*

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{x+1} + 8\times\frac{2x}{x^2+4} - \frac{1}{4}\frac{2x}{x^2+2}$$

*Finally, we multiply both sides by* $y$ *to obtain the derivative:*

$$\frac{dy}{dx} = \frac{(x+1)^3(x^2+4)^8}{\sqrt[4]{x^2+2}}\left[\frac{3}{x+1} + \frac{16x}{x^2+4} - \frac{x}{2(x^2+2)}\right]$$

*(e)* $x^{\sin y} = [\sin(x+y)]^y$

*Sol*

*By taking the* ln *for both sides, the given function can be written as:*

$$\sin y\ln x = y\ln(\sin(x+y))$$

*Now, we differentiate both sides:*

$$\frac{\sin y}{x} + y^{'}\cos(y)\ln x = y\frac{\cos(x+y)}{\sin(x+y)}\left(1+y^{'}\right)$$

*Therefore:*

$$y^{'} = \frac{y\cot(x+y) - \dfrac{\sin y}{x}}{\cos(y)\ln x - \cot(x+y)}$$

*(f)* $x = 2^y - 2^{-y}$

*Sol*

*Differentiate both sides directly:*

$$1 = 2^y \ln 2\, \frac{dy}{dx} + 2^{-y} \ln 2\, \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{(2^y + 2^{-y}) \ln 2}$$

**Exercise 1.2.1** *1. Find* $\frac{dy}{dx}$ *for the following:*

*(i)* $y = \pi^x - x^\pi$

*Sol*

*(ii)* $y = e^{\sin 3x}$

*Sol*

*(iii)* $y = \ln(x^2 - 1)$

*Sol*

*(iv)* $y = \ln\left(\frac{2^x - x}{2^x + x}\right)$

*Sol*

*(v)* $y = \log_3\left(\frac{x}{2^x - x}\right)$

*Sol*

*(vi)* $y = \tan^{-1}\left(e^{-x^2}\right)$

*Sol*

*(vii)* $y = \sin\sqrt{e^x - 1}$

*Sol*

*(viii)* $y = \dfrac{\ln(x) + 1}{\ln(x) - 1}$

*Sol*

*(ix)* $y = \dfrac{(x+1)^3\sqrt{x^2+4}}{\sqrt[3]{x+2}}$

*Sol*

*(x)* $y = \ln\left[\ln\left(\dfrac{x+1}{x-1}\right)\right]$



*Sol*

*(xi)* $y = \dfrac{\sin^4(x)\cos^3(x)}{\sqrt{x}}$

*Sol*

*(xii)* $xy = 2^{x-y}$

*Sol*

*(xiii)* $y + x = y^x$

*Sol*

2. *Find the following limits:*

   (a) $\lim_{x \to 1} \frac{\ln x}{x-1}$

   *Sol*

   (b) $\lim_{x \to 0} \frac{1-e^{2x}}{x}$

   *Sol*

   (c) $\lim_{x \to 0} \frac{e^{ax}-e^{bx}}{x}$

   *Sol*

   (d) $\lim_{x \to 0} \frac{e^{-x}-e^{2x}}{x}$

   *Sol*

   (e) $\lim_{x \to 0} \frac{e^{-x^2}-1}{x}$

   *Sol*